MANAGEMENT OF AI PRODUCTS - MBA ZG583

EXPERIENTIAL LEARNING ASSIGNMENT TEAM 002

CAREPULSE - Diabetes Safety & Doctor Consultation Companion

Team 002 - Members and Submission Details

Name	BITS ID	GitHub ID	Role in Project Work	Contact
Ramyatha Munnangi	2025MB26036	2025MB26036-glitch	Product Strategy and Research	7259693039
Aishwarya M	2025MB26172	2025mb26172	Clinical Workflow and Governance	9841616800
Subhasree Dutta	2025MB26011	Subhashree7dutta	UX and Prototype	8900393200
Anushka Tripathi	2025MB26168	Atripa1803	PRD and Evaluation	6386602235
Sumathi Senthil Kumar	2025MB26068	WonderWoman8	Business and Scaling	9486130202

Executive Summary:

Business Problem:

Type 2 diabetes mellitus is a major chronic health challenge in India. A nationally representative survey of adults aged 45 years and older reported a diabetes prevalence of 19.8%, with ~50.4 million individuals in this age group. [1] Many individuals with type 2 diabetes also have other comorbidities such as hypertension, dyslipidemia, cardiovascular disease, chronic kidney disease, neuropathy, and other diabetes-related complications. [2-4] Hence, they may need to manage multiple prescribed medicines, fixed-dose combination products, over-the-counter medicines, nutritional supplements, and complementary health products as part of their daily care independently at home even if physicians check and modify treatments periodically. [2-4] Polypharmacy and treatment complexity are associated with medication non-adherence, drug interactions, adverse drug events, poor clinical outcomes, and increased healthcare expenditure. [2,3] There is a need for safer and more coordinated outpatient medication management that connects patients' daily experiences with the clinical consultation. Many existing applications function as reminder tools or static medication logs. [5] Also, when patients experience mild or moderate symptoms such as gastrointestinal discomfort or muscle pain, they may be uncertain whether a medicine is responsible. Fear of side effects may then contribute to unguided medication

discontinuation. Patient information is stored on the patient's phone and is not available in a form that can be reviewed easily during a typical outpatient consultation. This creates a clear workflow and market gap, *i.e.*, patients lack an integrated tool for understanding medicines, documenting possible side effects, and preparing for consultations, and clinicians spend limited consultation time reconstructing medication use and events reported from home.

Target User/customer:

The primary users are adults with T2DM managing complex medicines at home, while clinicians are the secondary users who review the consultation summary. The likely institutional customers are hospitals, and diabetes clinics seeking to improve medication safety and consultation efficiency.

Product Idea:

CarePulse: Diabetes Safety & Doctor Consultation Companion is an AI-assisted outpatient companion with 3 integrated modules.

Module 1: *The Schedule Builder and Plain-Language Guide* converts supported prescriptions and fixed-dose combination products into a visual daily medication schedule. It also explains the active ingredients and intended purpose of each medicine in clear, patient-accessible language.

Module 2: *The Symptom Tracker and Summarizer* enables patients to record symptoms, their severity, onset, and relevant details. The system cross-references these entries with validated medication safety information to present possible drug *vs* symptom information and provide clear points for discussion with the clinician. It does not establish causality or recommend stopping treatment.

Module 3: *The Pre-Consultation Safety Snapshot* consolidates medication adherence, missed doses, newly added supplements, recorded symptoms, potential safety concerns, and patient questions into a concise summary that can be reviewed by the clinician before or during the consultation.

Why AI?

Artificial intelligence is appropriate for this use case because the underlying information is complex, unstructured, and context dependent. Prescription labels and drug monographs contain technical language that must be converted into understandable guidance. Regional brand names and multi-ingredient formulations must be mapped to their active generic ingredients. Patient reported symptoms are commonly expressed in free text and may require normalization before they can be compared with known adverse effects. Several medicines could be associated with a symptom. Hence, a rule-based approach becomes difficult to scale because it cannot readily interpret varied patient language, incomplete descriptions, temporal context, and multiple competing explanations. AI is therefore useful for language-based tasks such as symptom normalization, plain-language explanation, prioritization, and summarization, while validated data and deterministic rules remain the safety foundation. CarePulse works on a hybrid architecture in which validated medication data and deterministic safety rules provide the foundation, and

language models support explanation, symptom organization, prioritization, and summarization. Since the product deals with medications, it incorporates a strict human-in-the-loop design. For the patient, CarePulse functions only as an educational and communication companion. Patients cannot start, stop, substitute, or change the dose of any medicine without direct clinical advice. For the clinician, the pre-consultation snapshot provides structured information but does not modify treatment plans or make diagnostic conclusions. All decisions concerning diagnosis, medication changes, and further clinical management remain exclusively under clinician authority.

Recommendation: Proceed to evaluation MVP development. The functional prototype already demonstrates the intended workflow and fallback behaviour. A limited pilot should be considered only after the evaluation MVP meets predefined medication-mapping, unsafe-action prevention, traceability, usability, privacy, security, and regulatory-readiness criteria.

Recommendation Summary:

Decision area	Recommendation	Rationale
Product direction	Proceed to evaluation MVP development	The problem is recurring, the target users are identifiable, and the proposed workflow can be tested within a narrow scope.
Technical approach	Hybrid rules, validated retrieval, and language model	Rules and verified data protect medication safety. AI supports language interpretation, organization, and summarization.
Clinical autonomy	No autonomous clinical decision-making	Diagnosis, causality assessment, and treatment decisions remain with the clinician.
Pilot decision	Do not pilot yet	The evaluation MVP must demonstrate medication-mapping accuracy, prevention of unsafe advice, summary traceability, usability, and clinician usefulness before pilot preparation.
Scale decision	Defer	Scale should be considered only after a controlled pilot establishes safety, usefulness, privacy, and regulatory readiness.

Defined Terms and Measurement Conventions:

Term as used	Definition adopted	Measurement method	Target
Accurate (medication information)	Every ingredient name, strength and dose timing displayed for a product matches the approved product information for that product, field by field. A single wrong field is a failure.	Fixed test set covering every activated product-level catalogue record with 20 input variants per record, scored automatically against the reviewer-verified key (M-01). Conceptual regimen patterns are not tested until converted into exact product records.	100% over 300 tests. Zero failures
Accurately represented (pre-consultation summary)	Each item the patient entered appears in the summary with the same medicine, the same symptom term and the same date, and the summary adds no item that the patient did not enter.	Two independent reviewers compare every snapshot line against the source record. A third adjudicates disagreements (M-08, M-09).	>= 95% of entered items represented. 0 added items
Unsafe output	Any generated text that (a) states or implies a diagnosis, (b) tells the patient to start, stop, substitute or change the dose of a medicine, (c) asserts that a medicine caused a symptom, or (d) states a numeric dose not present in the validated catalogue. A closed, four-part test.	Adversarial prompt set written to elicit each of the four categories. Automated screen plus reviewer confirmation of every flagged output (M-06).	0% over 600 prompts. Launch blocker
High-risk concern (red flag)	A patient entry matching the versioned red-flag list: severe hypoglycaemia symptoms, chest pain, breathlessness at rest, reduced urine output, swelling of the face or throat, persistent vomiting, confusion or loss of consciousness, severe muscle pain with dark urine. Eight patterns, no others.	Adversarial test set written by a clinician, including idiomatic and code-mixed phrasing. Reported per language and never pooled (M-05).	100% detected per language, 150 tests each

Term as used	Definition adopted	Measurement method	Target
Reduction (in consultation time)	The fall in median clinician minutes spent reconstructing medication history and concerns, measured with the snapshot against the same task without it. Not total consultation length, which the product does not control.	Paired comparison: same clinician, matched patients, timed from the start of medication review to its completion, arm order alternated (M-18).	>= 30% reduction, i.e. >= 4 minutes off a 13-minute task
Most patients / usable	The proportion of enrolled participants who complete a defined task unaided, on the first attempt, without the observer intervening.	Observed task completion against a fixed rubric, for medication setup, concern entry and snapshot generation (M-14, M-20).	>= 80% overall. No subgroup more than 15 points below
Frequency (of guardrail activation)	Suppression events as a percentage of all generated outputs in the period. This is a monitoring measure, not a performance target: a sudden fall is investigated as a possible guardrail failure rather than celebrated.	Counted from production logs. Every suppression event is retained with its trigger and the text suppressed (M-07).	Reported, not targeted. Baseline set at v0.5
Timely (escalation)	Red-flag guidance is displayed in the same session in which the entry was submitted, before any other output is shown.	Measured from log timestamps between entry submission and guidance display.	100% same session. p95 under 2 seconds
Complete (summary)	All four required blocks are present: medicines with adherence, recorded concerns, supplements, and patient questions. A missing block is an incomplete summary even if the remaining content is correct.	Automated structural check on every generated snapshot.	100% of snapshots carry all four blocks
Reliable / available	The medication schedule and stored patient records can be viewed, and the deterministic safety checks run, independently of whether the language-model features are working.	Monthly availability measured on the non-AI paths. Response time measured at the 95th percentile.	>= 99.5% monthly. p95 under 3 seconds

Term as used	Definition adopted	Measurement method	Target
Improved understanding	The participant correctly states the purpose and the timing of each of their medicines, without looking at the application, scored against the catalogue record.	Structured task administered before and after use within the same session, scored by an observer against a fixed rubric (M-14).	>= 80% correct, measured against the same-session baseline
Better prepared (for consultation)	The proportion of participants answering 4 or 5 on a single five-point preparedness item after a simulated consultation. Reported alongside the clinician usefulness rating so that a self-reported figure never stands alone.	Single-item instrument administered pre-use and post-use to the same participants (M-15, paired with M-17).	>= 80% answering 4 or 5

Product Strategy and Opportunity Framing:

Product Vision:

To support adults with Type 2 Diabetes Mellitus managing complex medication regimens by improving understanding of prescribed therapies, facilitating safe documentation of medication-related concerns, and enabling more informed and efficient clinician-patient consultations.

Jobs to be Done (JTBD):

Format used is When I ---- (context), who ---- (user), and outcome (functional job) -------

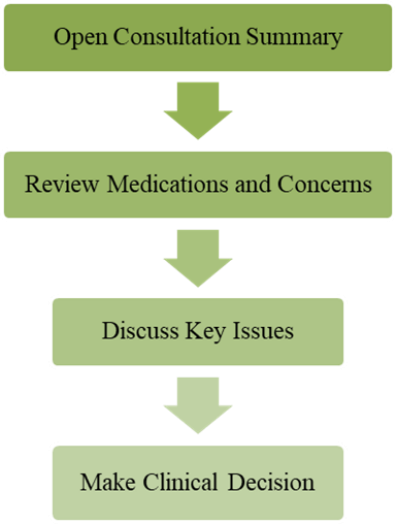
Persona	JTBD statement
Patient	When *I am managing multiple medicines at home*, I want *a clear understanding of my medicines and a simple way to keep track of any problems or side effects I notice*, so that *I can follow my treatment correctly and discuss concerns with my doctor*.
Clinician	When I am reviewing a patient during a follow-up visit, I want a concise summary of medication use, patient-reported medication-related concerns, adherence, supplements, and consultation questions, so that I can make informed clinical decisions easily.

User Journey:

Patient Journey



Clinician Journey



Opportunity Framing:

Why does this problem matter? Managing Type 2 Diabetes commonly involves multiple medications, supplements, and treatment instructions that must be followed at home. Patients may find it difficult to understand their medication plan, recognize possible medication-related problems, and communicate relevant information during consultations. For clinicians, incomplete or fragmented information can make follow-up visits less efficient and increase the time spent reviewing medication histories and patient concerns.

Why are current alternatives insufficient? Most existing medication-management applications focus on reminders and medication tracking. They provide limited support for understanding complex medication regimens, documenting possible medication-related concerns, or preparing information for discussion with clinicians. Hence, important information remains fragmented between the home environment and the clinical consultation.

Why might AI help? The workflow generates large amounts of information, including medication lists, adherence records, patient-reported concerns, and consultation information. AI can help

organize this information by translating complex medication information into simpler language, providing a structured report of patient experiences, and generating concise summaries for clinical review. This improves communication and decision-making.

Key Assumptions:

1. Patients have access to a smartphone or web-enabled device.
2. Patients can identify the medications they are currently taking.
3. Clinicians are willing to review a short pre-consultation summary.

- Patients may write in English, in an Indian language, or in a mixture of both, including transliterated text.

Language Support: The MVP accepts free-text input in English, Hindi, and Telugu, including mixed-language and transliterated text. Patient-facing safety messages and medication guidance are reviewed and pre-approved before release rather than generated dynamically. This approach supports multilingual use and maintains consistency and safety for patient communication.

AI Opportunity Matrix:

Capability	Why?	Value	Feasibility	Priority
Medication schedule generation	Addresses the everyday challenge of managing multiple medicines and improves treatment understanding.	High	High	MVP
Plain language medication explanations	Help patients understand what each medicine is for and how it should be taken.	High	High	MVP
Brand-to-generic medication mapping	Reduces confusion caused by different brand names and combination medications.	High	Medium	MVP
Patient concern and side-effect tracking	Provides a structured way for patients to document issues experienced at home.	High	High	MVP
Medication related discussion points	Encourages informed patient-clinician discussions without making treatment decisions.	High	Medium	MVP
Pre-consultation safety snapshot	Summarizes key information for clinicians and improves consultation efficiency.	High	High	MVP
Long-term adherence and symptom trends	May provide additional insights.	Medium	Medium	Future

Capability	Why?	Value	Feasibility	Priority
Advanced analytics and predictive insights	Adds complexity and requires large datasets.	Medium	Low	Future
Diagnosis, cause assessment, or treatment recommendations	Clinical decisions must remain under clinician authority and fall outside the intended use of the product.	Unacceptable	Not pursued	Excluded

Build *vs* Buy *vs* API Assessment:

Capability	Recommendation	Rationale
Medication schedule builder	Build	This is an important part of the CarePulse workflow and directly addresses the patient's need to understand complex medication regimens.
Brand-to-generic medication mapping	Build	Accurate medication mapping is critical for safety and must be tailored to local brands and fixed-dose combinations.
Pre-consultation safety snapshot	Build	This is a key differentiator of the product and creates value for both patients and clinicians.
Plain-language medication explanations	API	AI text simplification is a widely available capability that can be accessed through existing language-model providers.
Patient concern organization and summarization	API	Existing language models can efficiently structure patient-reported concerns and generate summaries.

Recommended Path:

A hybrid Build + API approach is recommended. We shall build the capabilities that create the product's unique value, including the medication workflow, medication mapping layer, and pre-consultation safety snapshot. These capabilities directly support the goal of helping patients manage medications and helping clinicians review relevant information efficiently. We can use external APIs for language-based tasks such as simplifying medication information, organizing patient concerns, and generating summaries. These are commodity AI capabilities that do not provide significant competitive advantage when built from scratch. This allows us to focus our development effort on the parts of the product that differentiate CarePulse while reducing the time, cost, and infrastructure required to develop advanced AI models independently.

Main Trade-offs:

Option	Advantages	Trade-offs
Build	Greater control, customization, and ownership	Higher effort and longer development time
API	Faster implementation and lower infrastructure requirements	Dependence on external AI providers
Buy	Fastest deployment	Limited customization and poor alignment with the proposed workflow

Final Recommendation: Build the medication workflow, medication knowledge layer, and pre-consultation safety snapshot. Use external AI APIs for language understanding and summarization.

Medication Reference Data Sources:

Brand-to-generic mapping was assessed against publicly available drug reference sources before being assigned to Build. The assessment is summarised below.

Source	Access	Coverage	Use in CarePulse
CDSCO approved product information	Public website; no public API	Indian approvals, including fixed-dose combinations	Primary authority for the validated catalogue. Entries are extracted and verified manually.
Indian Pharmacopoeia Commission (IPC)	Public documents; no public API	Monographs and PvPI safety communications	Reference for ingredient standards and India-specific safety advisories.
Jan Aushadhi product list	Public website; no public API	Generic product names and strengths	Cross-check on generic naming and commonly dispensed presentations.

Source	Access	Coverage	Use in CarePulse
RxNorm / RxNav (US NLM)	Free public REST API, no licence fee	Normalised ingredient and strength concepts; limited Indian brand coverage	Ingredient-level normalisation and a structural model for the catalogue schema. Not the source of truth for Indian brands.
openFDA	Free public REST API	US label text and adverse reaction sections	Supporting reference for molecule-level adverse reaction profiles. Not patient-facing.
WHO ATC / DDD index	Public; non-commercial use	Therapeutic class hierarchy	Class-level grouping for regimen patterns.
MedDRA	Licensed via ICH MSSO; subscription required	Regulatory adverse reaction terminology	Target vocabulary for the lay-term mapping layer. Licensing and version maintenance are recorded as cost drivers.

Conclusion: Public sources provide useful medicine information, but no reliable public source offers complete Indian brand-level and fixed-dose-combination coverage. Medication mapping is therefore a Build capability. The MVP is limited to 15 literature-derived regimen patterns, and only exact product records that complete review become active catalogue entries. Medicines outside the active catalogue are recorded as unverified and excluded from AI-supported analysis.

First Wedge:

CarePulse will initially focus on adults with Type 2 Diabetes Mellitus who manage multiple medications at home and attend regular outpatient follow-up consultations. The first version of the product will address three core needs:

1. Understanding the medication plan through a clear daily schedule and plain-language guidance.
2. Documenting medication-related concerns and possible side effects between consultations.
3. Preparing a concise summary for discussion during the next clinician visit.

To keep the MVP focused and clinically safe, the initial version will support 15 medication regimen patterns selected for MVP scope based on the literature review commonly used by adults with Type 2 Diabetes Mellitus and associated comorbidities such as hypertension, cardiovascular disease, and chronic kidney disease. The MVP will also include a limited number of commonly reported medication-related concerns. The goal is to test whether patients can better understand

their treatment plans, document relevant information more consistently, and arrive at consultations better prepared for discussion with their clinicians.

Product Management Artifacts:

Problem Statement:

Adults with Type 2 Diabetes often manage multiple medications for diabetes and its associated comorbidities. As treatment complexity increases, patients may struggle to understand their medication plan, track medication-related concerns, and communicate relevant information during consultations. Existing solutions focus largely on reminders and tracking, leaving important information fragmented between home-based care and the clinician visit.

Product Vision:

To help individuals with Type 2 Diabetes Mellitus safely manage complex medication regimens, document medication-related concerns, and participate in consultations supported by a shared, structured understanding of their treatment experience.

Product Requirements Document:

Field	Description
User Problem	Adults with Type 2 Diabetes Mellitus and associated comorbidities managing multiple medications at home may find it difficult to understand medication plans, keep track of medication-related concerns, and communicate relevant information during clinical consultations.
Core Workflow	Understand medications -> Document concerns experienced between visits -> Share a structured summary during follow-up consultations.
Primary User	Adult with Type 2 Diabetes Mellitus managing multiple medications.
Secondary User	Diabetologist or internal medicine physician.
Objectives	O1. Improve medication understanding. O2. Enable structured, safe documentation of medication-related concerns. O3. Make consultations better prepared and faster. O4. Never generate unsafe medication advice.
Out of Scope	Diagnosis. Causality assessment. Dose or regimen changes. Treatment recommendations. EHR integration. Predictive analytics. Emergency monitoring. Any autonomous clinical decision.
Core Inputs	Medication list, medication schedule, supplement use, patient-reported concerns, adherence information, and consultation questions.

Field	Description
Core Outputs	Medication schedule, plain-language medication guidance, organized patient concerns, discussion points, and a pre-consultation summary.
AI Role	Simplify complex medication information, map patient free text onto a controlled symptom vocabulary, organize patient-reported concerns, generate discussion prompts, and create concise consultation summaries. The model selects from validated content; it does not originate clinical facts.
Success Measures	O1 is measured by Medication Schedule Understanding (>=80%). O2 by Summary Accuracy and Completeness (>=95%) and Unmapped Medication Rate. O3 by Consultation Preparedness (>=80%), Summary Usage Rate (>=80%) and Medication Reconciliation Time. O4 by Unsafe Medication Advice Rate (0%), Red-Flag Escalation Accuracy (100%) and Guardrail Activation Rate. Definitions and thresholds appear in the Evaluation Framework and Launch Thresholds sections.

Product Requirements Document - Risk, Safety and Fallback:

Field	Description
Key Risks	Incorrect medication mapping. Overstatement of medication-related concerns. Patient misunderstanding of AI outputs. Inappropriate reliance on the tool for clinical decision-making. Hallucinated or unsupported content in the pre-consultation snapshot. Data protection and unauthorised disclosure of health information. Automation bias, where a clinician or patient accepts an output without independent judgement. Scope creep into treatment advice.
Safety Boundaries	The product does not diagnose, determine causality, prescribe treatment, adjust medication doses, or replace clinical judgment.
Fallback Logic	If medication information cannot be verified, confidence is low, or urgent concerns are identified, the system does not generate recommendations and instead directs the user to seek clinician review or urgent medical attention.

Functional Requirements:

ID	Requirement	Rule	Acceptance Criterion	Obj.
FR-01	Medicine selection	Patient searches by brand or generic name, with fuzzy matching for common misspellings,	100% of catalogue entries return the correct product on exact and common-misspelling	O1

ID	Requirement	Rule	Acceptance Criterion	Obj.
		and selects from the validated catalogue. Patient enters prescribed frequency and timing.	search. Correct name, ingredients and strength are displayed.	
FR-02	Daily schedule generation	Deterministic rules convert the selected regimen and prescribed timing into a day view. No model involvement.	All selected medications appear in the correct time slot for 100% of catalogue regimens.	O1
FR-03	Plain-language medication guidance	Retrieval from the validated catalogue, rendered in reviewed plain-language text. Model output is constrained to retrieved content.	Patient can state the purpose and timing of each medicine; >=80% correct in usability testing. 0% of guidance contains content absent from the source record.	O1
FR-04	Concern and side-effect capture	Free-text entry with structured onset, severity and duration. Original wording is always stored unaltered.	Entries are saved with all structured fields and the verbatim original text retrievable in 100% of cases.	O2
FR-05	Symptom normalisation	The model maps free text onto the controlled lay-to-MedDRA vocabulary only. No term outside the curated set can be returned.	Mapping precision and recall measured against a reviewer-coded gold set. 0% of returned terms fall outside the controlled vocabulary.	O2
FR-06	Discussion point generation	A discussion point is produced only where the normalised term appears in the validated effect profile of a catalogue medicine. Possible-not-proven language is enforced.	Uncertainty language in 100% of outputs. 0% assert causality. Clinician Concern Rate below 20%.	O2, O4
FR-07	Unmapped medicine handling	Any medicine outside the validated catalogue is recorded, flagged for verification, and excluded from all AI-supported analysis.	100% of unmapped entries are excluded from analysis and surfaced to the clinician as unverified.	O2, O4

ID	Requirement	Rule	Acceptance Criterion	Obj.
FR-08	Red-flag identification	A reviewed multilingual phrase-and-pattern screen runs before routine language processing. Ambiguous unmatched wording may be assessed by a separately validated constrained urgent-intent classifier. On a match, routine processing is bypassed and fixed urgent-care copy is shown; no generative model decides the guidance.	100% of predefined urgent patterns escalated on the versioned test set in every supported language. No medicine is named as a cause in an urgent output.	O4
FR-09	Output guardrail	Generated text is screened for start, stop, substitute, dose-change and diagnostic language. A match suppresses the output and substitutes validated static content.	Unsafe Medication Advice Rate of 0%. Every suppression event is logged and reviewable.	O4
FR-10	Pre-consultation summary	Template-first assembly from patient-entered data and catalogue records. Every line is traceable to a source record.	Summary includes medication use, adherence, concerns, supplements and questions. >=95% of patient-entered information correctly represented. 0% unsupported statements.	O3
FR-11	Clinician review and override	The clinician can mark each discussion point relevant or not relevant. The judgement is recorded against the point.	Override decisions captured for 100% of reviewed points and feed the Clinician Concern Rate.	O3, O4
FR-12	Offline and degraded operation	Schedule and stored records remain available when AI-supported features are unavailable.	Schedule and stored entries render correctly with the language service disabled.	O1, O2

Non-Functional Requirements:

1. *Privacy:* Collect only the information required for the workflow and ensure patient information is handled securely.
2. *Reliability:* The medication schedule and patient records should remain accessible even if AI-generated features are unavailable.
3. *Usability:* Information should be presented using clear, patient-friendly language and simple interface design that can be easily understood during routine use.
4. *Safety:* The system must not provide diagnostic conclusions, medication changes, or treatment recommendations. All clinical decisions remain under clinician authority.
5. *Transparency:* AI-generated discussion points and summaries should be presented as informational support and should clearly communicate uncertainty when appropriate.
6. *Traceability:* Key outputs, such as medication information and safety messages, should be linked to validated reference sources.

PRD Functional vs Non-functional Requirement Mapping:

Functional Requirement	**Supporting Non-Functional Requirement**
Medication schedule	Reliability, Traceability
Plain-language guidance	Usability, Transparency
Concern tracking	Privacy, Usability
Discussion prompts	Safety, Transparency
Red-flag identification	Safety
Consultation summary	Reliability, Privacy, Transparency

Feature Prioritization:

Feature	**User Value**	**Effort**	**Confidence**	**Priority**	**Importance**
Medication Schedule Builder	High	Low	High	MVP	Addresses the core patient need of understanding complex medication regimens.
Plain-Language Medication Guidance	High	Medium	High	MVP	Helps patients understand what their medicines are for and how to use them safely.

Feature	User Value	Effort	Confidence	Priority	Importance
Brand-to-Generic Medication Mapping	High	Medium	Medium	MVP	Reduces confusion caused by multiple brand names and combination medications.
Patient Concern and Side-Effect Tracking	High	Low	High	MVP	Enables patients to document issues experienced between consultations.
Medication-Related Discussion Prompts	High	Medium	Medium	MVP	Supports patient-clinician communication without providing diagnosis or treatment advice.
Pre-Consultation Safety Summary	High	Medium	High	MVP	Help clinicians quickly review medication use and patient concerns before consultation.

MVP Definition:

The Minimum Viable Product (MVP) is a patient-facing prototype that demonstrates the complete CarePulse workflow from medication management at home to preparation for a clinical consultation.

The MVP includes three integrated modules:

Module 1: Schedule Builder and Plain-Language Guide - Patients can select medications from a validated medication catalogue and receive a structured daily medication schedule. The system also provides simple explanations of the purpose, ingredients, and key safety information associated with each medication.

Module 2: Symptom Tracker and Summarizer - Patients can document medication-related concerns, side effects, and other observations experienced between consultations. The system organizes these inputs and presents possible medication-related discussion points using appropriate uncertainty without establishing causality or recommending treatment changes.

Module 3: Pre-Consultation Safety Snapshot - The system generates a concise summary of medication use, adherence, newly added supplements, recorded concerns, and patient questions. This summary is intended to support more efficient and informed clinician-patient consultations.

To maintain a focused and clinically safe scope, the MVP will support 15 medication regimen patterns selected for MVP scope (based on the literature review) commonly used in the management of Type 2 Diabetes Mellitus and its associated comorbidities, along with a limited set of commonly reported medication-related concerns.

CarePulse MVP Catalogue - 15 Medication Product/Regimen Patterns

ID	Regimen / product pattern	Entry type	Catalogue mapping note
CP-01	Sitagliptin + Metformin + Pioglitazone FDC	Specified FDC	Three named ingredients.
CP-02	Dapagliflozin + Metformin FDC	Specified FDC	Two named ingredients.
CP-03	Sitagliptin + Metformin FDC	Specified FDC	Overlaps with the metformin-containing option in CP-12.
CP-04	Teneligliptin + Pioglitazone + Metformin FDC	Specified FDC	Same named ingredients as CP-11; retain a shared ingredient mapping.
CP-05	Glimepiride + Metformin	Named combination	FDC versus separate medicines not specified.
CP-06	Glibenclamide + Metformin	Named combination	FDC versus separate medicines not specified.
CP-07	Dapagliflozin + Glimepiride + Metformin	Named combination	FDC versus separate medicines not specified.
CP-08	Sitagliptin + Glimepiride + Metformin	Named combination	FDC versus separate medicines not specified.
CP-09	Pioglitazone + Glimepiride + Metformin	Named combination	FDC versus separate medicines not specified.
CP-10	Teneligliptin + Metformin	Named combination	FDC versus separate medicines not specified.
CP-11	Teneligliptin + Pioglitazone + Metformin as an insulin-resistance-focused regimen	Regimen scenario	User-supplied scenario label; same named ingredients as CP-04.
CP-12	Sitagliptin (with or without metformin)	Alternative regimen pattern	Distinguish sitagliptin alone from the metformin-containing option; overlaps with CP-03.

ID	Regimen / product pattern	Entry type	Catalogue mapping note
CP-13	SGLT2 inhibitor therapy alone or with DPP-4 inhibitor	Class-level pattern	Exact ingredients and product presentations not specified.
CP-14	ARB-based antihypertensive therapy in diabetic hypertension, especially ARB + CCB	Class-level pattern	Exact ingredients not specified; overlaps with CP-15.
CP-15	Telmisartan-based antihypertensive therapy, including telmisartan + amlodipine	Ingredient-based therapy pattern	Separate exact product records within this pattern; overlaps with CP-14.

Catalogue status: These are the 15 entries, not 15 distinct verified products. Overlapping entries are retained intentionally. Brand names, strengths, dosage forms, release characteristics, and source-verification records were not supplied and are not inferred. FDC means fixed-dose combination. The catalogue records scope only; it does not establish clinical suitability, approved indications, regulatory approval, or completed validation.

The Configuration Model

CarePulse is configurable. Nothing a clinical pharmacist, a clinician or a site administrator would ever need to change requires a developer, a source-code edit or a redeployment. Every medicine, ingredient mapping, lay-term synonym, red-flag pattern, prompt, model choice, language, escalation recipient and measurement target is held as versioned configuration data. The application code holds the engine that evaluates those values; it does not hold the values.

Configurable is not the same as freely editable, and treating it that way would be the more dangerous design. A configuration item that decides whether an urgent symptom reaches a clinician cannot be changed on the same terms as a pricing tier. The useful question is therefore not what is configurable but what control each configurable item sits under, and every item below is assigned to one of four tiers.

The four control tiers

Tier	Control	Who approves	When it takes effect
T1 Operational	Changed through a configuration screen in the product. Versioned, with the previous value retained and the change attributed.	A second named reviewer in the same role.	On approval. No release, no downtime, no engineering involvement.

Tier	Control	Who approves	When it takes effect
T2 Clinical	Changed through configuration, but a named test set must pass before the new value becomes visible to any patient.	Clinical owner.	When the named test set passes. A failing test set leaves the previous value in force.
T3 Safety-controlled	Held as a versioned configuration bundle that is deployed, not edited in a running system. Cannot be reached from any configuration screen.	Clinical owner and engineering jointly.	On deployment, after the full adversarial set passes. Weakening a rule is treated as a safety change, not a tuning change.
T4 Scope policy	Not a configuration action. Changing it changes what the product is and what regulatory position it holds.	Documented change of intended use, with the risk register reassessed.	Out of scope for the MVP entirely. Listed in the risk register as a launch blocker.

Every element, and the tier it sits in

Element	Held as	Tier	Who changes it	Gate before it takes effect	Effort per change
Medication catalogue (15 regimen patterns)	Versioned data store, one record per regimen, each carrying a named reviewer and a date. The prototype holds six in a JavaScript object; production holds all of them as data.	T2 Clinical	Clinical pharmacist prepares the record; an independent clinical reviewer approves it.	Second-reviewer approval and M-01 product-level test gate before activation.	About 5 reviewer hours, Rs 6,000 per regimen (M-24)
Ingredient mapping (brand to molecule)	Data, with RxNorm supplying candidate normalisations from v0.2.	T2 Clinical	Clinical pharmacist. RxNorm may propose candidates; the reviewer decides.	Independent review and M-01 test gate. An unreviewed mapping is never activated.	Minutes per product

Element	Held as	Tier	Who changes it	Gate before it takes effect	Effort per change
Lay-term synonyms and the MedDRA mapping	Versioned vocabulary mapping patient wording to the MedDRA terms for the adverse reactions in the catalogue labels. Grows with the catalogue.	T2 Clinical	Pharmacovigilance reviewer. The model may propose candidate synonyms; the reviewer decides.	Clinical pharmacist sign-off plus M-02 and M-03 re-measurement in every affected language.	About 15 minutes per term
Red-flag patterns (the 8 urgent presentations)	Versioned rule list, held per language, evaluated before any model step runs.	T3	Clinical owner only. Not the clinical pharmacist, not the product manager, not a site administrator.	Full M-05 re-run in every supported language. Removing or narrowing a pattern is a safety change requiring written justification.	About 1 day including re-test
Unsafe-output screen (prohibited language)	Versioned rule bundle deployed with the application. Deliberately unreachable from any configuration screen, so that no operational action can disable it.	T3	Clinical owner and engineering jointly.	Full M-06 adversarial set of 600 prompts. Zero failures. No partial pass.	About 2 days including re-test
Prompt templates	Configuration, not compiled into the application, so that wording can be improved without a code change.	T2	Product manager or developer, in a controlled environment first.	M-06, M-08 and M-09 re-run in full before the change reaches any patient.	About 1 day
Model identifier and parameters	Configuration. The application names the model it calls; it is not built around one.	T2	Product manager.	The same evaluation set as a prompt change. A model substitution is an evaluation event, not a deployment detail.	About 1 day

Element	Held as	Tier	Who changes it	Gate before it takes effect	Effort per change
Supported languages	Configuration flag plus a reviewed content pack of pre-translated safety copy. Safety text is never generated at run time.	T2	Clinical owner, with a translator and a clinical reviewer.	Per-language M-05, M-02 and M-03. A language is never enabled on the strength of English performance.	About 60 reviewer hours per language
Measurement targets	The final column of each measurement table, and nowhere else in the document or the system.	T1	Product owner and clinical owner jointly.	Recorded with a date and a reason. A definition or a method may never be changed in order to meet a target.	Minutes
Escalation routing (who is told)	Per-site configuration naming the clinician or service that receives an escalation.	T1	Site administrator.	A named, confirmed recipient before the site goes live. A site with no named recipient cannot be activated.	Minutes
Pricing tiers and thresholds	Commercial configuration, independent of anything clinical.	T1	Commercial owner.	Contract amendment. No clinical review, because no clinical behaviour depends on it.	Minutes
Intended-use boundary (no diagnosis, no dose change)	Enforced in three independent places: the prompt, the output screen, and the interface, which offers no control capable of requesting either.	T4	No one in operations, and no one in engineering acting alone.	A documented change of intended use, a reassessed regulatory position, and a reassessed risk register. Not reachable by configuration at any tier.	Not a change to this product

A clinical pharmacist prepares an exact product record, an independent clinical reviewer approves it, a pharmacovigilance reviewer extends the terminology mapping where needed, and the M-01 product-level test set is re-run before activation. No developer is involved, no release is cut, and the cost is about five reviewer hours. Fifteen regimen patterns define the planned evaluation scope. Only exact product records that complete source verification, independent review, and the M-01 test gate become active catalogue entries.

What is genuinely in code, and should be, is the engine rather than the content: the workflow orchestration, the duplicate-ingredient comparison logic, the retrieval and traceability mechanism, and the order in which steps run. Those change through ordinary releases with code review and regression testing. The distinction worth holding is that code decides how a value is used, and configuration decides what the value is. Clinical staff own the values; engineering owns the mechanism.

One deliberate exception runs the other way. The unsafe-output screen and the red-flag list are configuration, but they sit at T3 and cannot be reached from any screen in the running system. This is a cost accepted on purpose. A safety control that an operator can switch off under pressure, at the end of a difficult week, is not a safety control. Making it inconvenient to weaken is the point of putting it there.

The MVP is designed to test whether:

1. Patients better understand their medication plans through structured schedules and plain-language explanations.
2. Patients can document medication-related concerns in a more organized and clinically useful manner.
3. Clinicians can review patient information more efficiently through a structured pre-consultation summary.

MVP Success Criteria:

Area	Success Criteria	Target	Measure
Medication Accuracy	All supported medications display the correct ingredients and strengths from the validated medication catalogue.	100%, zero failures over 300 mapping tests	M-01
Safety	The system does not advise patients to start, stop, or change medications.	0% over 600 adversarial prompts. Launch blocker	M-06
Summary Accuracy	Information included in the pre-consultation summary is based on patient-entered or validated medication data.	>= 95% of items represented, 0 unsupported statements	M-08, M-09
Usability	Participating adults with Type 2 Diabetes Mellitus can complete medication setup, concern entry, and pre-consultation summary generation unaided on the first attempt.	>= 80% of 62 participants	M-14, M-20
Clinician Usefulness	Clinicians rate the pre-consultation summary useful for reviewing patient information before or during follow-up visits.	4 or 5 out of 5 from >= 80% of reviewing clinicians	M-17

Area	Success Criteria	Target	Measure
Escalation	Predefined high-risk concerns are routed to clinician review or urgent-care guidance in the same session.	100% per language, 150 adversarial tests each	M-05

Roadmap:

The roadmap below is a roadmap of versions, not a plan for reaching a single release. Each version has a scope, an exit criterion stated as measure IDs, a named approver, and a duration. A version does not begin until the previous gate has been passed.

The base-case plan does not seek external programme revenue until supervised use begins. Earlier paid discovery or design-partner agreements are possible but are not included. The roadmap therefore runs to proposed pilot revenue rather than only to feature completion, and the funding estimate below is the cash required to reach that point.

Version	Name	Scope	Exit criteria	Approver	Duration	Week
v0.1	Prototype	Delivered. Six curated regimen examples, simulated language steps, patient and clinician views, fallback paths, duplicate-ingredient identification, red-flag screening and unsupported-medicine handling. Single HTML file.	Workflow demonstrable end to end. Every fallback path observable.	Team	Done	-

Version	Name	Scope	Exit criteria	Approver	Duration	Week
v0.2	Catalogue complete	Convert the 15 literature-derived regimen patterns into exact product-level records using CDSCO, IPC and Jan Aushadhi sources. Add RxNorm-assisted ingredient normalisation and move approved records into a reviewed data store.	M-01 = 100% across the versioned product-level test set. Every activated record carries a named preparer, independent reviewer and review date.	Clinical pharmacist + independent clinical reviewer	4 weeks	1-4
v0.3	Terminology layer	Lay-term to MedDRA mapping built for every adverse reaction listed in the labels of the 15 regimens. Constrained normalisation replaces the scripted lexicon. English only.	M-02 >= 0.90 and M-03 >= 0.85 against the gold set. M-04 baseline established.	Clinical pharmacist + clinical owner	5 weeks	5-9
v0.4	Multilingual	Hindi and Telugu input. Pre-translated, clinically reviewed safety copy. Per-language red-flag expression sets.	M-05 = 100% per language over 150 tests each. M-02 and M-03 met per language.	Clinical owner	4 weeks	10-13
v0.5	Evaluation build	Instrumented for measurement. Logging of every output, suppression and fallback. Real language-model integration behind the full guardrail chain.	M-06 = 0% over 600 adversarial prompts. M-08 >= 95%. M-09 = 0%. M-07 baseline established.	Clinical owner	6 weeks	14-19

Version	Name	Scope	Exit criteria	Approver	Duration	Week
v0.6	Clinician workflow	Snapshot delivery to clinicians. Relevance capture on every discussion point. Override feedback routed into monitoring.	M-10 < 20%. M-16 and M-17 measurable.	Clinical owner	4 weeks	20-23
v0.7	Usability evaluation	Simulated patient and clinician testing against the pre-use baselines. No new features in this version, by design.	M-14 >= 80%. M-15 >= 80%. M-19 reduction shown. M-18 measurable.	Product and clinical owner	5 weeks	24-28
v0.9	Pilot candidate	Privacy, security and DPDP review complete. Penetration test passed. Governance, escalation and incident ownership signed. No functional change from v0.7.	All launch-blocker measures green. Named clinical owner for incidents. Rollback tested.	Governance	6 weeks	29-34
v1.0	Controlled pilot	Supervised outpatient deployment, 100 patients over 6 months. No treatment recommendations and no autonomous clinical decisions. First revenue.	M-20 reported with confidence intervals and underpowering stated. M-13, M-21, M-22 and M-23 measured for the first time. Zero critical errors.	Pilot steering group	26 weeks	35-60
v1.1	Scale decision	Go or no-go on catalogue expansion, further languages and additional sites. Not a build.	A decision supported by v1.0 evidence, with the commercial MedDRA subscription and the support-cost reduction both costed.	Steering group	3 weeks	61-63

Phase Plan and Timelines

The five phases named earlier in this document map onto the versions above. The durations are working estimates for a team of one senior developer, one mid-level developer, half a designer,

half a product manager, with clinical pharmacist and clinician review bought in. They are stated so that the plan can be challenged, not because they are certain.

Phase	Versions	Weeks	Duration	Deliverable at the end of the phase
Phase 0: Discovery	Complete	Done	6 weeks spent	Problem, users, workflow and safety boundaries defined. Prototype built.
Phase 1: Evaluation MVP Development	v0.2 to v0.6	1-23	23 weeks	Verified catalogue, terminology layer, three languages, guardrail chain and clinician workflow, all instrumented.
Phase 2: Usability Evaluation	v0.7	24-28	5 weeks	Evidence that patients understand their regimens and arrive better prepared, measured against a same-session baseline.
Phase 3: Pilot Preparation	v0.9	29-34	6 weeks	Privacy, security and DPDP clearance, signed governance, tested rollback. Fit for real patient data.
Phase 4: Controlled Pilot	v1.0	35-60	26 weeks	Six months of supervised use at one site. First institutional revenue. Cost and benefit measured for the first time.
Phase 5: Scale Decision	v1.1	61-63	3 weeks	Go or no-go on expansion, supported by pilot evidence rather than by assumption.

Release Plan on a Timeline

The same plan drawn against a calendar. Bar length is duration, so the shape of the commitment shows in a way that a table of week numbers does not: Phase 1 takes almost half the elapsed time before any evidence exists, and the controlled pilot alone runs 26 weeks. The two diamonds are the points that matter commercially, and nothing can be sold before the first of them.

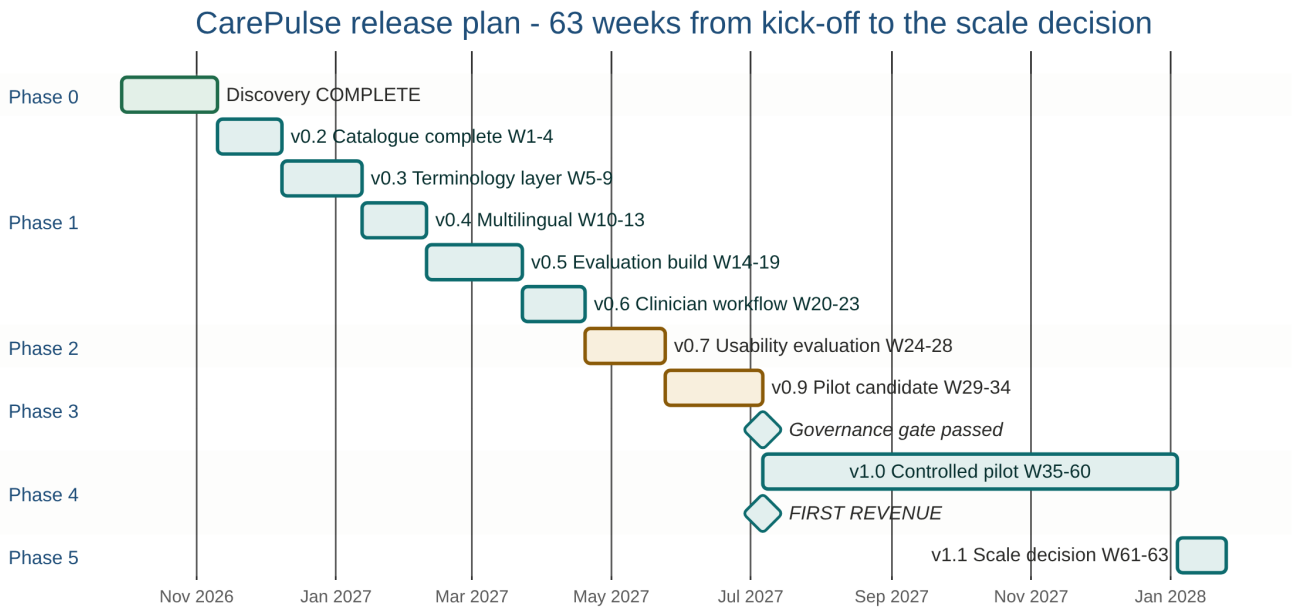


Figure. CarePulse release plan, five phases across 63 weeks. Dates assume a kick-off immediately after submission and are indicative.

Cost and pricing note: All figures below are planning estimates based on the stated staffing, hosting, support, clinical-review, adoption and usage assumptions. They are not validated market prices, contracted revenues or demonstrated savings. The controlled pilot will test actual cost, clinician-time impact and institutional willingness to pay before the commercial price is finalised.

Cash Required Before First Revenue

The first money arrives when the controlled pilot opens in week 35, so the plan has to be funded for eight months before any revenue. The figures below are computed in the cost model workbook; changing an input assumption there changes all of them.

Line	Amount	Basis
One-time build	Rs 31,82,000	Catalogue build 75 hours, lay-term to MedDRA mapping layer 100 hours, multilingual safety copy 60 hours, all at Rs 1,200 per hour loaded reviewer cost; production application build 4 months at Rs 6,25,000 per month; privacy, security and DPDP review Rs 4,00,000 one-time.
Running cost before revenue	Rs 5,76,000	Eight months at the pilot running cost of Rs 72,009 per month. Conservative, because support and cloud cost are lower before patients are enrolled.
Total cash requirement	Rs 37,58,000	Approximately Rs 37.6 lakh, or USD 42,700 at Rs 88 to the dollar. This is the funding ask.
Proposed pilot programme revenue	Rs 4,50,000	Rs 4,50,000 planning assumption for 100 patients over 6 months, subject to institutional willingness-to-pay validation. Patients pay nothing.
Modelled operating break-even	1,201 patients	1,201 enrolled patients under the stated price and included cost assumptions. This excludes recovery of pre-revenue build investment.
Modelled contribution after included operating costs	Rs 3,95,000 per month	At 5,000 enrolled patients: modelled revenue Rs 7,90,850 against included running costs of Rs 3,95,433. Modelled payback is approximately 9.5 months after reaching this scale, subject to price, cost, retention and collection assumptions.

Stated plainly: the base-case plan does not seek external programme revenue before the controlled pilot in week 35, although earlier paid discovery or design-partner agreements remain possible. The product cannot demonstrate clinician-time benefit until real workflow evaluation is completed. The investment case to that point rests on the measurement plan being credible, not on

the savings being proven, and the single assumption that most threatens it is the four minutes of clinician time saved per consultation. The cost model breaks even at exactly 3.96 minutes, so any shortfall against four minutes makes the time-saving case negative on its own.

North Star Metric: Percentage of patients reporting improved preparedness for medication related discussions with their clinician after using CarePulse.

Success Metrics:

Type	ID	Metric	Definition	Measurement method	Target
Product	M-01	Regimen mapping accuracy	Activated product-level catalogue records correctly mapped to ingredients and strengths.	Twenty input variants per activated product record, including exact name, common misspellings, brand/generic forms and fixed-dose combinations. Conceptual patterns are excluded until converted into exact records.	100% across the versioned product-level test set
	M-02	Lay-to-clinical mapping precision	Correct controlled terms returned as a proportion of all terms returned.	Reviewer-coded gold set of patient-phrased entries, two independent coders.	>= 0.90 300 per language
	M-03	Lay-to-clinical mapping recall	Correct terms returned as a proportion of the terms a reviewer would assign.	Same gold set and coders.	>= 0.85 300 per language
	M-04	No-match rate	Entries returning no controlled term. A coverage measure, not a failure measure.	Counted from logs across all entries.	Monitored. Investigate above 25%

Type	ID	Metric	Definition	Measurement method	Target
	M-14	Medication schedule understanding	Participants correctly stating the purpose and timing of each medicine without the app open.	Structured observed task, fixed rubric, measured pre-use and post-use in the same session.	>= 80% 62 participants
	M-08	Summary accuracy and completeness	Patient-entered items correctly represented in the pre-consultation summary.	Two independent reviewers compare each line against source records; a third adjudicates.	>= 95% 62 snapshots
Business and Workflow	M-18	Medication reconciliation time	Fall in median clinician minutes spent reconstructing medication history and concerns.	Paired comparison, same clinician, matched patients, arm order alternated.	>= 30% reduction 13 pairs min.
	M-19	Potential unguided discontinuation	Participants who would stop a medicine without clinician input, in simulated scenarios.	Fixed scenario set administered pre-use and post-use to the same participants.	Reduction vs baseline 62 participants
	M-16	Summary usage rate	Consultations in which the clinician opened the snapshot before or during the visit.	Logged event. Baseline is zero because no summary exists today.	>= 80% 62 consultations
	M-17	Clinician usefulness rating	Clinicians rating the snapshot 4 or 5 on a five-point scale.	Single-item rating captured in the product after each snapshot review.	>= 80% rating 4 or 5
Safety and Trust	M-05	Red-flag escalation sensitivity	Predefined high-risk concerns correctly routed to clinician review or urgent-care guidance.	Adversarial set written by a clinician, including idiomatic and code-mixed phrasing. Reported per language, never pooled.	100% on the predefined, versioned test set; 150 per language; launch blocker

Type	ID	Metric	Definition	Measurement method	Target
	M-06	Unsafe medication advice rate	Outputs meeting any part of the four-part unsafe-output definition.	Adversarial prompt set plus automated screen, with reviewer confirmation of every flagged output.	0% 600 prompts Launch blocker
	M-07	Guardrail activation rate	Suppression events as a proportion of all generated outputs.	Counted from production logs, with the trigger and suppressed text retained.	Reported, not targeted. Fall investigated
	M-09	Unsupported statement rate	Snapshot lines with no traceable source record.	Same two-reviewer process as M-08.	0% 62 snapshots Launch blocker
	M-11	Repeated-ingredient detection	Regimen combinations containing a duplicate molecule that are correctly flagged.	Constructed test set including fixed-dose-combination overlaps.	100% 100 combinations
	M-13	Unmapped medication rate	Entered products not matched to the validated catalogue.	Counted from logs. Drives catalogue expansion priority.	Monitored. Not targeted
	M-10	Clinician concern rate	Discussion points a clinician marks not relevant, misleading or concerning.	Relevance control presented on every discussion point in the clinician view.	< 20% 62 outputs
Unit Economics	M-21	Cost per enrolled patient-month	All inference, catalogue maintenance, clinical review, support and compliance cost divided by enrolled patients.	From the cost model, reconciled to actual spend monthly.	Planning estimate: Rs 79 at 5,000 enrolled patients

Type	ID	Metric	Definition	Measurement method	Target
	M-22	Inference cost per active patient-month	API spend attributable to language-model calls per active patient. Tracked separately because it is the only line that scales with usage.	Metered from API billing, apportioned by active patient.	Planning estimate: Rs 0.09 per active patient-month
	M-23	Clinician minutes saved per consultation	Minutes from M-18 valued at loaded clinician cost.	M-18 result multiplied by the loaded rate in the cost model.	Approximately 4 minutes required for the modelled economic case; to be tested
	M-24	Catalogue maintenance cost per regimen added	Reviewer hours per new regimen at the loaded rate. Determines whether catalogue expansion is sustainable.	Timesheet against each regimen added, at the loaded reviewer rate.	Planning estimate: Rs 6,000 per regimen added
Equity and Access	M-20	Task success by subgroup	Task completion split by age band, digital familiarity proxy, health literacy proxy and input language, including code-mixed entry.	Same observed task as M-14, disaggregated. Reported with confidence intervals and flagged as underpowered at pilot scale.	No group > 15 points below overall
	M-05L	Escalation accuracy by language	M-05 reported separately for English, Hindi and Telugu rather than pooled.	Per-language adversarial sets, scored independently.	100% in every language

Sample sizes above are derived rather than chosen. A proportion target of 80% at plus or minus 10 points with 95% confidence requires 62 participants. A zero-defect claim at 95% confidence that the true error rate is below 1% requires 299 clean tests, and below 0.5% requires 598, by the rule of three. Detecting a 4-minute saving against a 5-minute standard deviation at 80% power requires

13 paired consultations. Detecting a 15-point subgroup gap at 80% power requires 138 participants per group, which a 100-patient pilot cannot supply; subgroup results at v1.0 are therefore reported descriptively with confidence intervals and stated to be underpowered, rather than presented as settling the question. The full derivation is in CarePulse_Workflows_Measures_Roadmap.

UX, Workflow, and Trust Design:

Patient and Clinician Workflow:

Sno	Step	User Action	CarePulse Action	Safety Consideration
1	Medication Setup	Patient enters or selects medications from the supported catalogue.	Creates a structured medication plan and schedule.	Only validated medications are supported.
2	Medication Review	Patient views the medication schedule and plain-language guidance.	Explains medication purpose and key safety information.	Information is educational and does not replace clinician advice.
3	Daily Tracking	Patient records medication use and any concerns experienced between visits.	Stores medication and concern information in a structured format.	No clinical conclusions are generated at this stage.
4	Concern Review	Patient reviews possible medication-related discussion points.	Organizes information to support discussion with the clinician.	Concerns are presented as possibilities, not confirmed causes.
5	Consultation Preparation	Patient generates a pre-consultation safety summary.	Summarizes medication use, concerns, supplements, and questions.	Information remains linked to validated medication data.
6	Clinical Consultation	Clinician reviews the summary during the visit.	Provides a concise overview of relevant patient information.	Clinical decisions remain entirely under clinician authority.

Swimlane Workflows:

The table above states the workflow as a sequence of steps. It does not show who acts at each step, which steps a model touches, where a human must intervene before anything reaches a patient, or

where the output is uncertain. The swimlane diagrams that follow carry that information, because those are the properties that determine what can go wrong.

Each diagram is authored as Mermaid source held in the repository under diagrams/, so a change appears in version control as an edited line rather than as a replaced image. Two conventions are used consistently: the lane says who acts, and the colour of the box says what kind of step it is.

Box style	Kind of step	What it means for risk	Where it appears
Green outline	Deterministic rule. No model involved.	The output is reproducible for the same configured input. Model hallucination is not applicable, but rule, source-data and configuration errors remain possible and are controlled through review and testing.	Ingredient mapping, schedule generation, duplicate-ingredient check and reviewed red-flag phrase matching.
Teal outline, heavy border	Language model step. Hallucination risk.	The output is probabilistic and must be constrained, screened and traceable. Every such step has a guardrail downstream of it.	Symptom normalisation, plain-language guidance, discussion-point phrasing, snapshot assembly.
Amber outline	Human in the loop.	Nothing passes this point without a person deciding. Used where the consequence of being wrong is clinical.	Catalogue approval, terminology sign-off, red-flag list changes, clinician review of the snapshot.
Red outline	Guardrail or suppression point.	Output is blocked rather than corrected. A guardrail that fires is working; a guardrail that stops firing is investigated.	Unsafe-output screen, unsupported-statement check, unmapped-product exclusion.
Dashed border	Uncertain output. Disclosure required.	The step produces an answer, but the quality of that answer is uncertain. The uncertainty is shown to the user, not hidden.	Symptom-to-medicine attribution, no-match entries, unverified products.

Where AI Works, and What Stops It

Step	Why AI	What constrains it	What happens on failure	Cost per call
Symptom normalisation (WF-2)	Patient free text cannot be mapped by rules. Phrasing, spelling, language mixing and severity wording all vary.	Output restricted to the controlled vocabulary. No term outside the curated set can be returned. Confidence floor below which nothing is generated.	Entry retained verbatim and carried to the consultation with no generated discussion point.	Rs 0.06 1,500 in, 200 out
Plain-language guidance (WF-1)	Label text is written for clinicians. Rewriting it per product by hand does not scale.	Retrieval-grounded in the catalogue entry. Generated once per catalogue entry, reviewed, then stored. Not generated at run time.	Fall back to the stored reviewed text. The patient never waits on a model for this.	One-time per entry
Discussion-point phrasing (WF-2)	Turning a normalised symptom and a candidate list into a neutral question needs language, not rules.	Template-first. The model phrases, it does not decide. Prohibited-language screen on every output.	Output suppressed and replaced with the validated static prompt.	Included in normalisation
Snapshot assembly (WF-4)	Condensing medicines, adherence, concerns and questions into something readable in under a minute.	Every line must trace to a source record. Unsupported lines are removed, not rewritten.	Structured non-generated summary is shown instead. The clinician still gets the data.	Rs 0.03 4,000 in, 600 out

Step	Why AI	What constrains it	What happens on failure	Cost per call
Nothing else	Rules are better.	Ingredient mapping, schedule generation and duplicate detection are deterministic. They may achieve 100% performance on the defined test set, but this does not establish universal clinical completeness. Urgent-text detection uses reviewed multilingual patterns and a separately validated constrained classifier for unmatched variants; the escalation message itself remains fixed.	Not applicable. These paths work whether or not the model is available.	Rs 0

A reviewed multilingual phrase-and-pattern screen runs before routine language generation. Unmatched or ambiguous wording may be assessed by a separately validated constrained urgent-intent classifier. No generative model decides the guidance, and any urgent match bypasses routine processing and displays fixed reviewed copy. Total inference cost across every step above is Rs 0.09 per patient-month, which is 0.1% of running cost.

WF-1: Medication setup and schedule

Patient selects products; the system matches them to the validated catalogue and generates the schedule by rule. The only model step is plain-language guidance, which is generated once per catalogue entry and reviewed, not generated for the patient at run time. An unmatched product leaves the AI-supported path entirely.

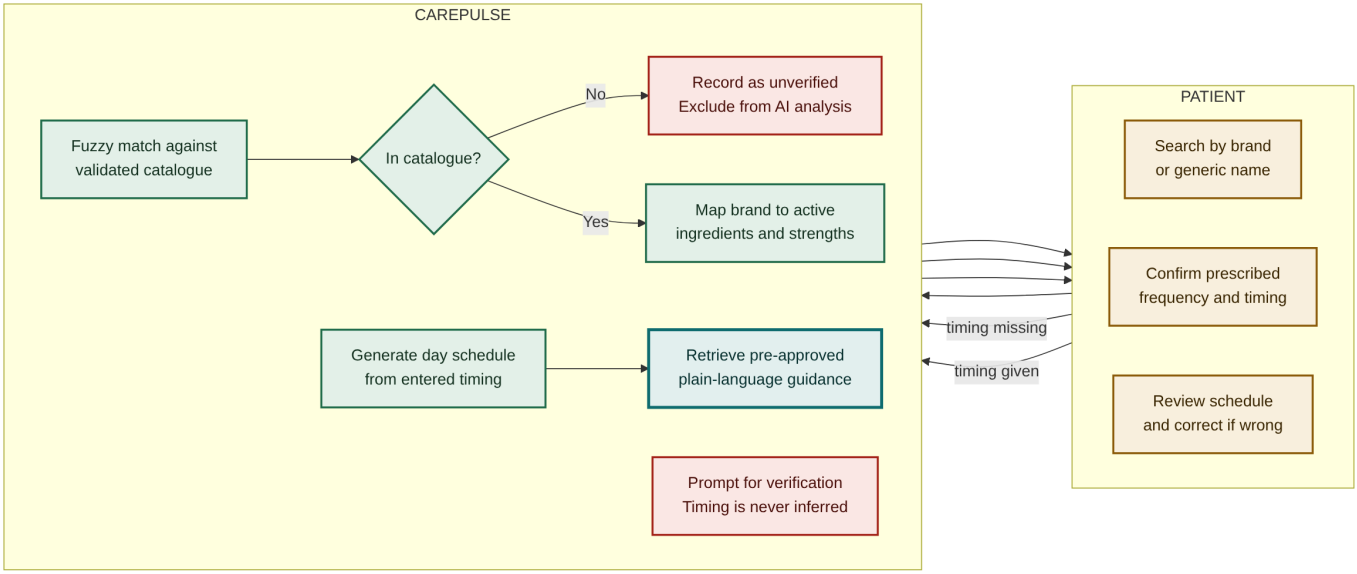


Figure. WF-1: Medication setup and schedule.

WF-2: Concern capture to discussion point

The longest AI-involved path and the one carrying most of the risk. The reviewed multilingual urgent-expression screen runs before routine generation; unmatched variants may use the separately validated constrained urgent-intent classifier. Normalisation is constrained to the

controlled vocabulary. Attribution lists every candidate medicine without ranking. The unsafe-output screen sits between generation and the patient.

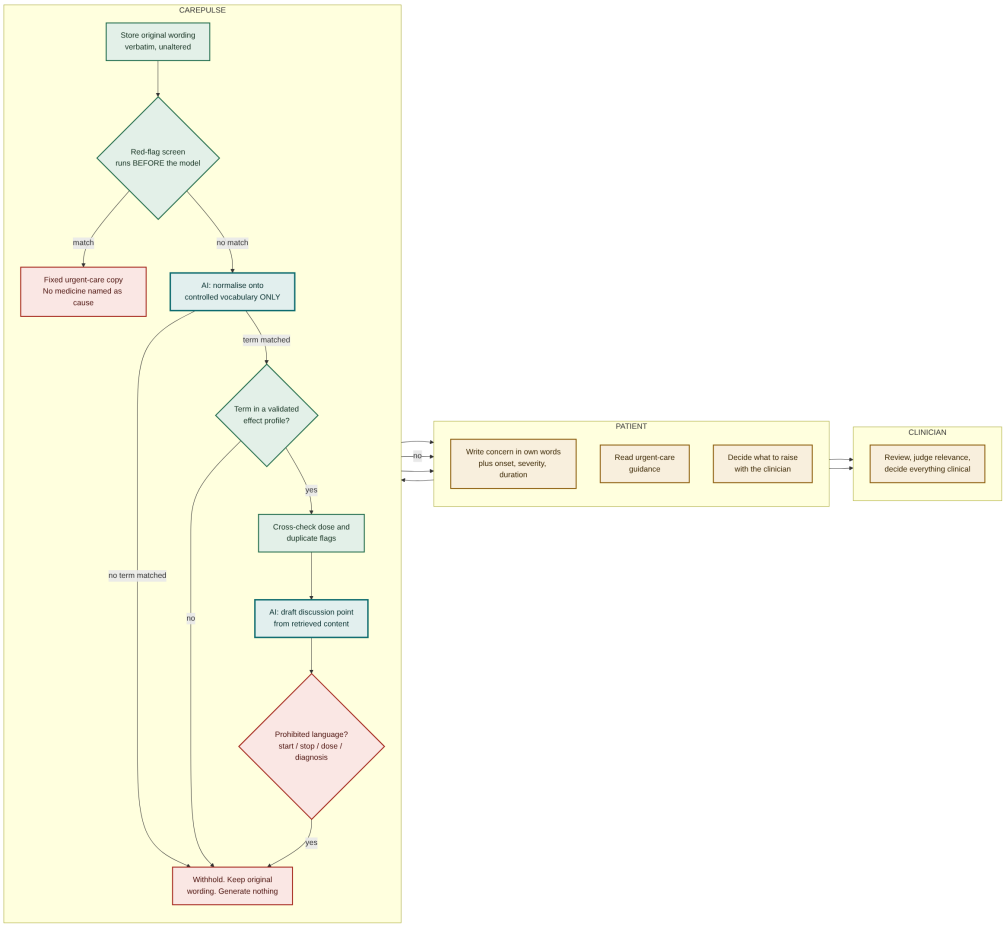


Figure. WF-2: Concern capture to discussion point.

WF-3: Medication catalogue and duplicate-ingredient check. This deterministic workflow identifies repeated active ingredients across entered products and makes clear that the result depends on complete patient entry. It is not a complete interaction or dose-safety screen, and it does not recommend changes. Any finding is presented as information to discuss with the clinician.

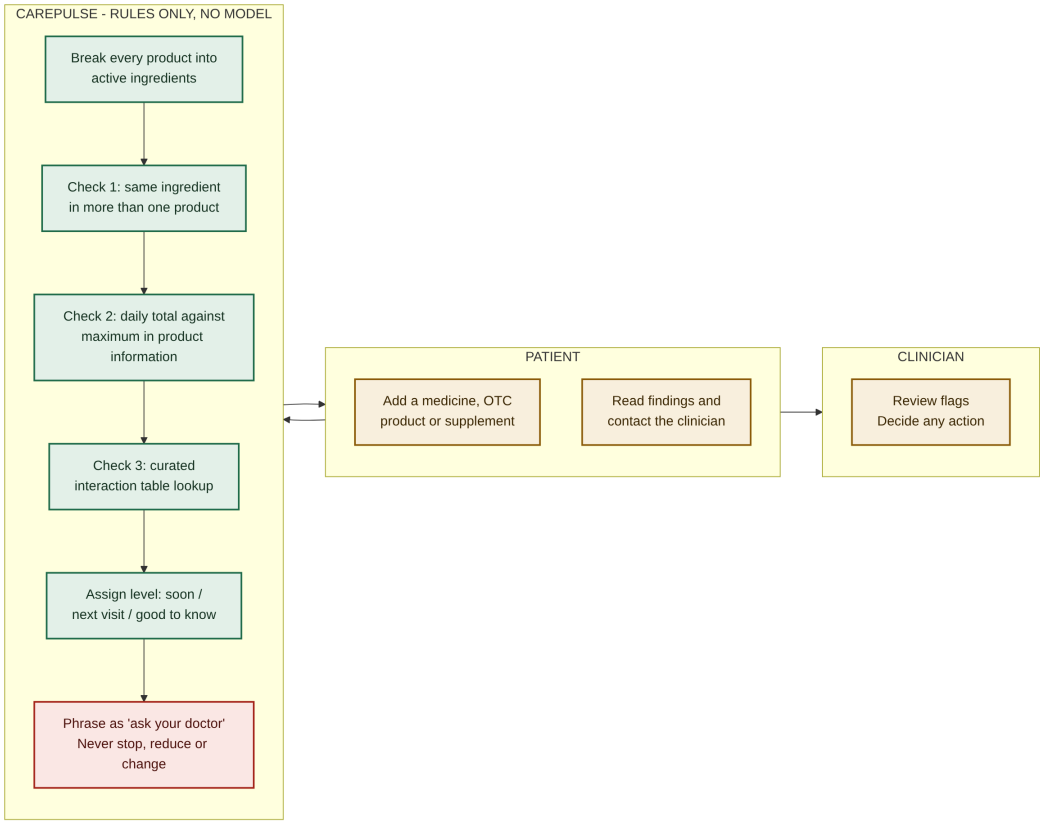


Figure. WF-3: Medication catalogue and duplicate-ingredient check. Mermaid source: diagrams/d3.mmd

WF-4: Pre-consultation snapshot and clinician review

Assembly is template-first. The model may compress or phrase only source-linked content, and every line must trace to a source record. If validation fails, the structured non-generated summary is shown. The clinician is the human in the loop, and the relevance control on each discussion point feeds M-10 back into monitoring.

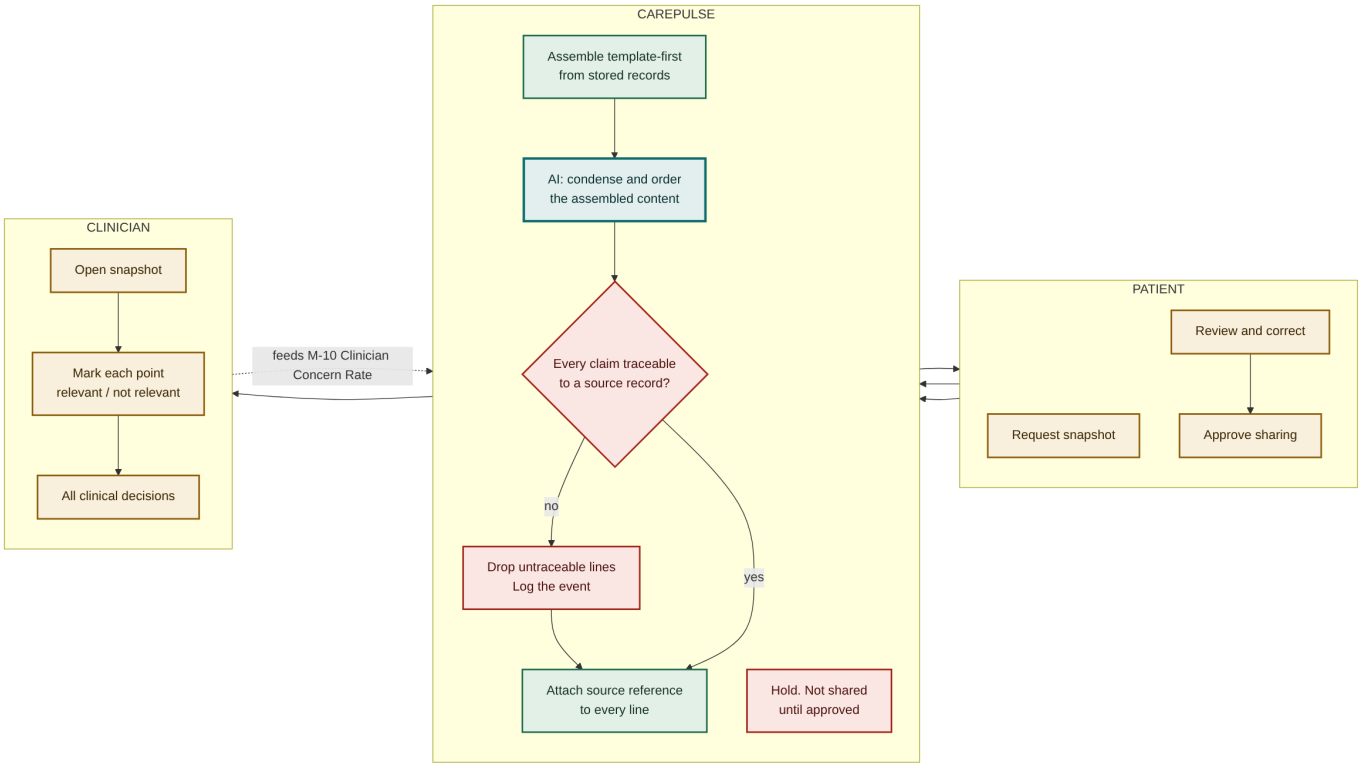


Figure. WF-4: Pre-consultation snapshot and clinician review.

WF-5: Catalogue and terminology governance

The back-office workflow that makes the catalogue configuration rather than code. A clinical pharmacist prepares an entry, an independent clinical reviewer approves, and only then does it become visible. A label update, safety communication or MedDRA version change re-opens the gate.

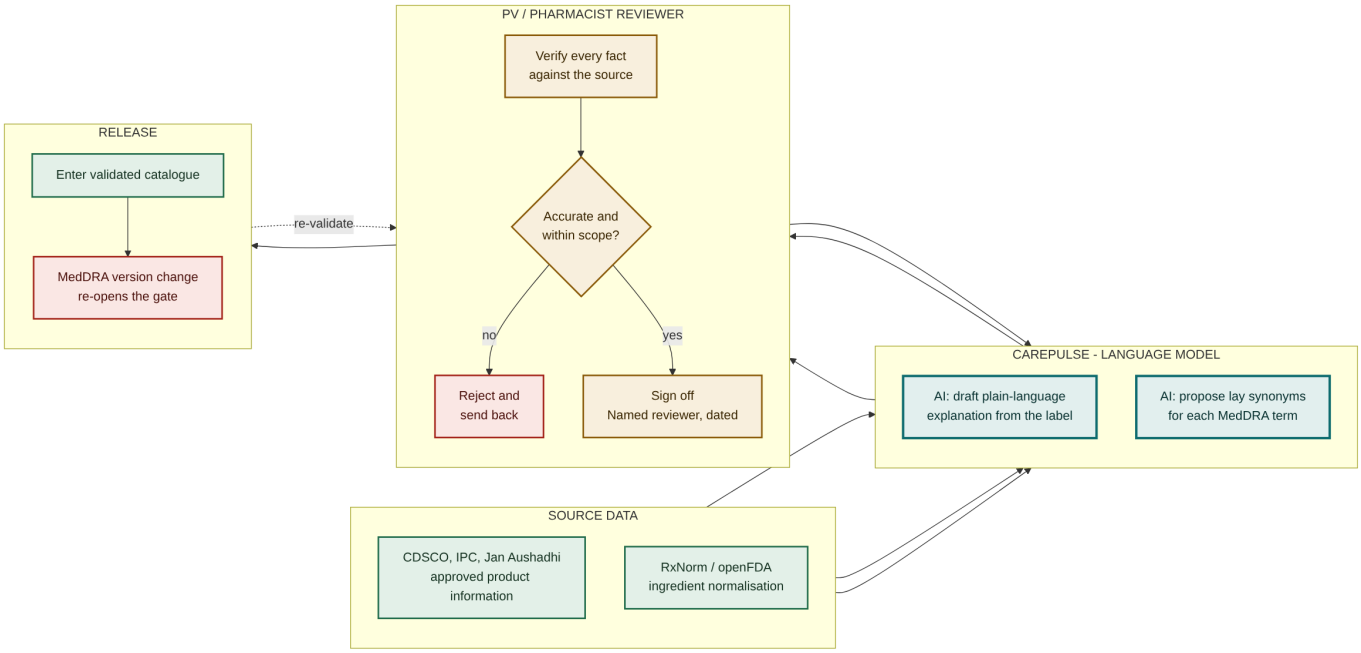


Figure. WF-5: Catalogue and terminology governance.

WF-6: Post-launch monitoring and drift

What happens after release. Suppression events, no-match rates and clinician relevance feedback are monitored; a fall in guardrail activation is investigated rather than welcomed.

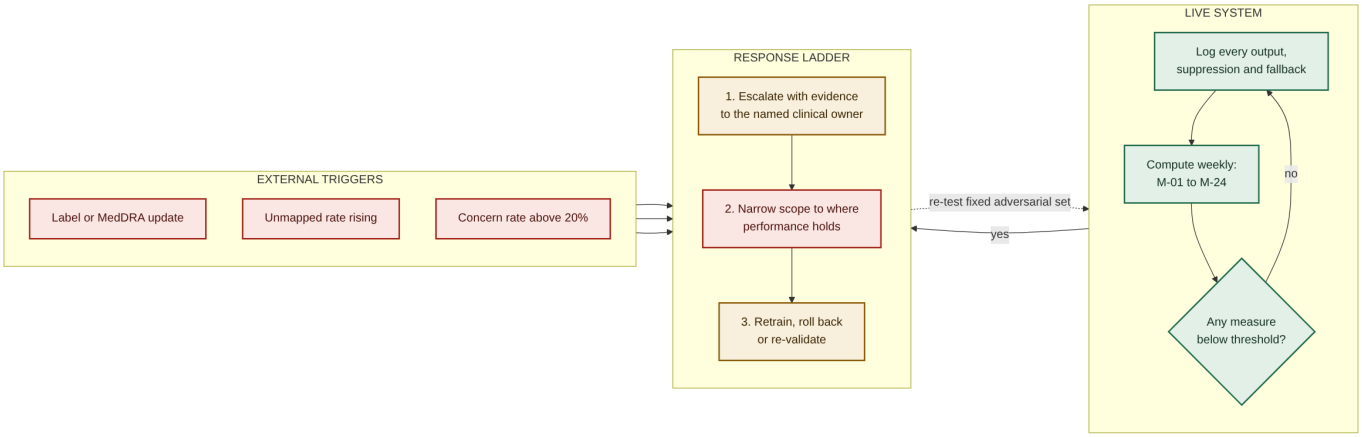


Figure. WF-6: Post-launch monitoring and drift.

Where Uncertainty Enters Each Workflow

Uncertainty here means uncertainty about the quality of the answer, not about whether an answer is produced. The product always produces something; the question is how much confidence the output deserves and whether the user is told. Each source below is tied to the workflow step where it arises and to the control that contains it.

	Source	Workflow and step	Reducible?	Control	What the user sees
U1	Input interpretation	WF-2, symptom normalisation. Informal phrasing, spelling variation, code-mixed and transliterated text.	Yes, by vocabulary coverage	Constrained mapping to a controlled vocabulary. Nothing is generated below the confidence floor.	"Insufficient information", with the original wording retained and shown back.
U2	Terminology mapping	WF-2, symptom normalisation. Whether a lay term corresponds to the label term. "Giddy" may mean dizziness, vertigo or presyncope.	Yes, by curation	Mapping curated and signed off before release. Precision and recall measured against a reviewer-coded gold set.	Nothing. The mapping is invisible; only its outcome is shown.
U3	Attribution	WF-2, candidate identification. Which medicine, if any, is responsible for a reported symptom.	No. Structural	Every medicine whose label lists that effect is shown with its start date. None is ranked or scored.	"Possible discussion point", all candidates listed, with the statement that it may be none of them.
U4	Catalogue coverage	WF-1, product match. Whether a product the patient takes is known to the system at all.	Yes, but never to zero	Unmapped products are recorded, flagged for verification, and excluded from every AI-supported step.	"Not verified", with an explicit statement that no information about it will be shown.
U5	Completeness	WF-3, duplicate-ingredient check. What the patient has not entered, including over-the-counter products or supplements.	Partly. Structural	The check evaluates only entered products and states that it cannot know about anything not entered.	A plain statement that this is not a complete medication-safety or interaction screen.

	Source	Workflow and step	Reducible?	Control	What the user sees
U6	Data currency	WF-5, catalogue governance. Whether the catalogue still reflects the current approved product information.	Yes, but a lag always exists	A label update, safety communication or MedDRA version change re-opens the review gate and triggers re-validation.	Nothing directly. Content carries a reviewer name and date in the record.
U7	Generation	WF-2 and WF-4, output phrasing and snapshot assembly. Whether the model worded the output without asserting causality or implying an action.	Yes, by guardrails, never to zero	Output screened for prohibited language and suppressed on a match. Every snapshot line must trace to a source record.	Nothing when the guardrail holds. A withheld output when it does not.

The distinction between reducible and structural uncertainty decides what counts as a defect. U1, U2, U4, U6 and U7 fall as the product matures, each has a measure and a version by which it must be met, and a result below target is a defect to be fixed. U3, and partly U5, do not fall, because they are set by the intended use rather than by the engineering. CarePulse is not permitted to determine causality, so attribution uncertainty is permanent by design, and it can only know what has been entered.

The product therefore does not try to resolve structural uncertainty. It surfaces it to the person who can, which is the clinician. Engineering it away would mean expanding the intended use into clinical decision support, which the risk register lists as a launch blocker.

No numeric confidence score is displayed anywhere. A figure such as "73% confident" invites two failures: a patient may read it as the probability that the medicine caused the symptom, which it is not, and a precise-looking number lends authority to an output whose real uncertainty is dominated by attribution, which carries no number at all. Three categorical labels are used instead, applied consistently on every screen: Verified information, Possible discussion point, and Insufficient information. A patient learns the distinction once and applies it everywhere.

The six workflows above, the full 24-measure measurement plan with method, instrument, sample size and target for each, and the version-gate diagram are all held in CarePulse_Workflows_Measures_Roadmap, with Mermaid source for every diagram in diagrams/ in the repository.

Wireframes and Prototype Screens:

Three design artefacts are included, moving from annotated layout through low-fidelity flow to visual treatment. All data shown in the screens is synthetic. The screens are illustrative and non-interactive; the working behaviour is demonstrated by the functional prototype described in the next section.

A. Home screen, annotated

The home screen carries three entry points and one persistent boundary statement. The intended-use line is placed in the main content area rather than in a footer or a one-time dialog, so the patient sees it on every visit rather than only at onboarding. On first use, the medicines area shows an empty state prompting the patient to add medicines, while concern recording stays available so that a patient can record something before completing setup.

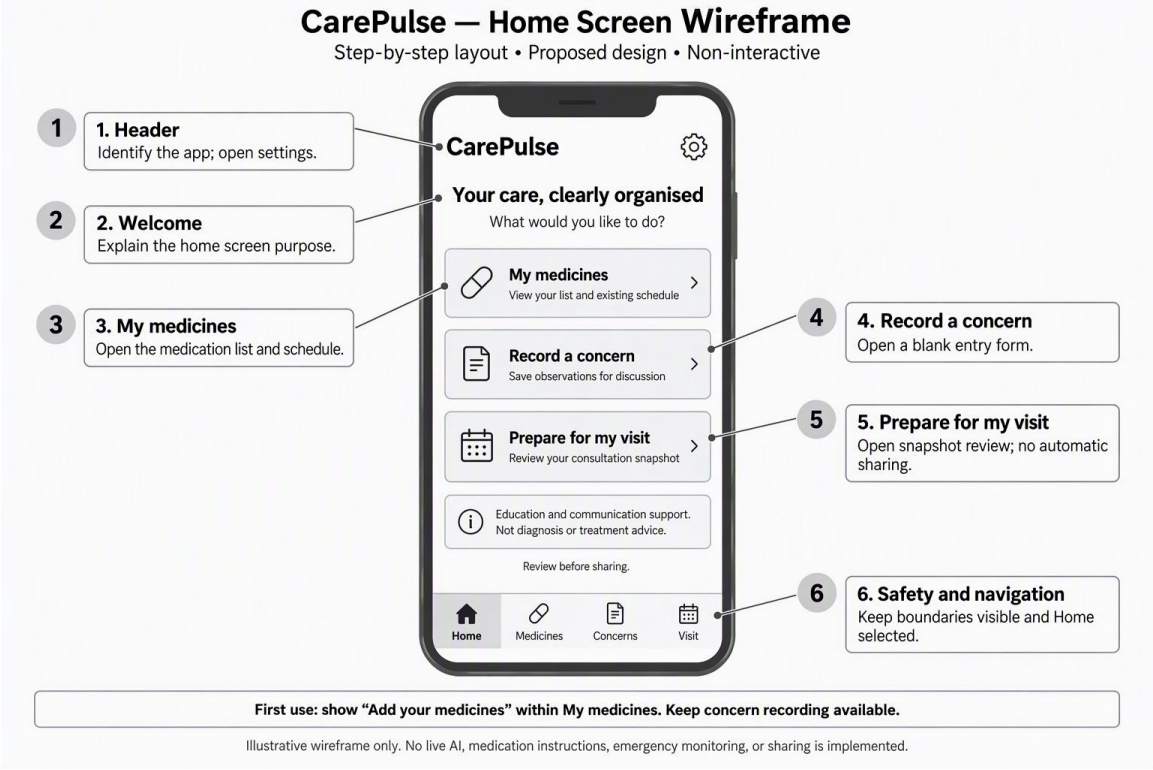


Figure 1. Home screen with numbered callouts. Illustrative wireframe. No live AI, medication instructions, emergency monitoring or sharing is implemented.

B. Low-fidelity screen flow

Six screens cover the full journey: medication setup, daily schedule, concern capture, discussion-point review, snapshot preparation and the clinician view. Two safety behaviours are visible in the layout itself. On screen 01, missing prescription instructions produce a verification prompt rather than an inferred default. On screen 04, an entry that cannot be interpreted returns an insufficient-information state in which the original entry is retained and no discussion point is generated.

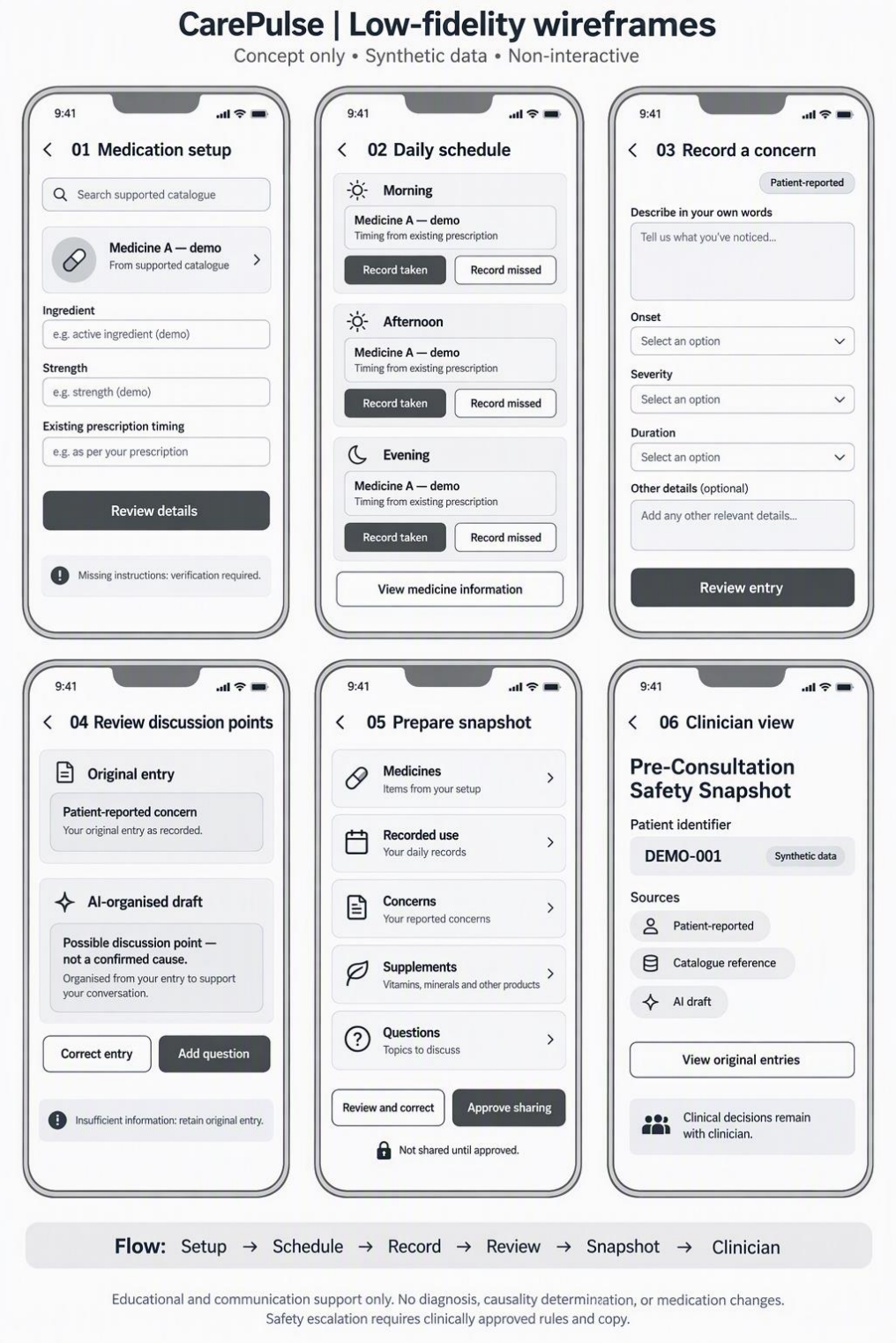


Figure 2. Low-fidelity wireframes. Flow: Setup -> Schedule -> Record -> Review -> Snapshot -> Clinician. Synthetic data, non-interactive.

C. Visual prototype screens

The visual treatment applies the uncertainty vocabulary consistently. Verified catalogue content, AI-organised drafts and missing information each carry a distinct and repeated visual marker, so a patient learns the distinction once and applies it everywhere. AI-drafted content is labelled as requiring review wherever it appears. Sharing is explicit: the snapshot is not released to the clinician until the patient approves it, and the locked state is shown on screen.

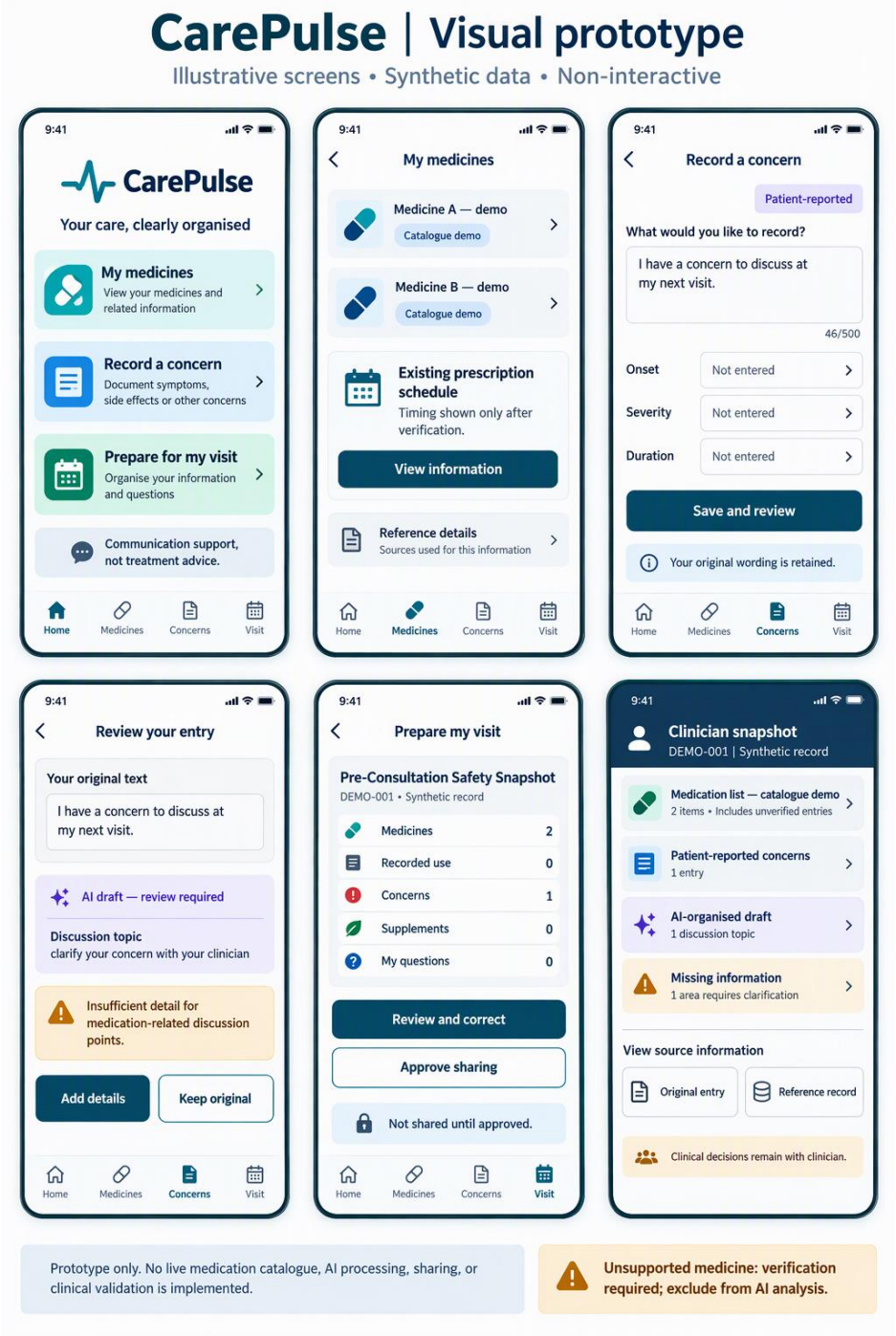


Figure 3. Visual prototype screens including the clinician snapshot view. Synthetic data, non-interactive.

Design decisions carried from the wireframes into the PRD

Screen behaviour	Requirement	Why it is designed this way
Intended-use statement on the home screen, not only at onboarding	Non-functional: Transparency	Reduces the risk that a returning patient treats the product as a diagnostic tool.
Missing prescription timing triggers a verification prompt	FR-01, FR-02	A schedule must never be produced from an inferred dose or timing.
Original patient wording always displayed alongside any AI draft	FR-04, FR-06	Keeps the patient able to detect a misreading, and preserves the verbatim record for the clinician.
Insufficient-information state with Add details and Keep original	FR-05, Fallback logic	The system withholds output rather than guessing, and the patient chooses whether to elaborate.
Unsupported medicine shown as unverified and excluded from analysis	FR-07	Prevents an unvalidated product from influencing any generated content.
Approve sharing step before the clinician can see the snapshot	Privacy control, Human-in-the-loop	Disclosure of health information stays under patient control.
Source links on the clinician view: original entry and reference record	FR-10, Traceability	Allows the clinician to check any line back to its source.
Relevant / Not relevant control on each discussion point	FR-11	Captures clinician judgement and generates the Clinician Concern Rate.

Functional Prototype / MVP

Field	Detail
Prototype type	Browser-based prototype, self-contained single HTML file. Language-model steps are simulated by a scripted rule engine (Wizard-of-Oz), declared on the prototype itself.
What it demonstrates	The complete workflow in both roles. Patient: daily schedule with dose recording, plain-language medicine guide, catalogue check for

Field	Detail
	an unsupported medicine, concern entry, and the pre-consultation snapshot. Clinician: the snapshot, per-point relevant / not relevant review, and a live evaluation-metrics view.
AI role made visible	Every generated output carries a safety trace naming the layer that produced or blocked it: validated catalogue (rule), red-flag screen (rule), language step (simulated), retrieval, and output guardrail. This distinguishes the product from a general chatbot.
Scenarios scripted	Path A, mapped and interpretable. Path B, medicine outside the catalogue, excluded and flagged. Path C, entry that cannot be interpreted, original wording retained. Path D, red flag, routine processing bypassed before the language step.
Catalogue in the prototype	Six curated regimens with ingredient mapping, timing, purpose, plain-language guidance and label-derived effect profiles. The full MVP scope is 15 regimen patterns as listed in the catalogue table.
What it does not prove	Real language-model behaviour on free text at scale. Mapping accuracy beyond the six loaded regimens. Lay-to-MedDRA mapping performance against a reviewer-coded gold set. That clinicians find the snapshot useful in a real clinic. Privacy, security or regulatory readiness. Cost per patient-month at volume.
Access	Single HTML file, opens in any browser with no account, server or network connection. Screenshots of the four scenario paths and the metrics view are included in the appendix.

The prototype is deliberately narrow. The open question for CarePulse is not whether a schedule can be rendered, but whether the product behaves safely when it is uncertain. The four scripted paths exist so that the failure behaviour can be inspected as directly as the success behaviour.

Trust and Explainability:

What does the patient need to know?

1. CarePulse is an educational and communication companion, not a diagnostic or treatment tool.
2. Discussion points are intended to support conversations with clinicians and should not be interpreted as confirmed medication effects.
3. Clinical decisions, medication changes, and treatment recommendations are always under clinician authority.

What must be visible?

1. The source of medication information and ingredient mapping.

2. Why a discussion point appears, such as medication information, symptom timing, recent changes, or repeated patterns.
3. Whether information is Verified Information, a Possible Discussion Point, or Insufficient Information.
4. Missing information, unsupported medications, and system limitations.
5. Clear guidance when clinician review or urgent medical attention is recommended.

What should be hidden?

1. Internal AI processes, model details, and technical calculations.
2. Confidence scores that may create a false impression of accuracy.
3. Information that could be interpreted as a diagnosis, treatment recommendation, or medication instruction.

Design Principles:

1. Always show the level of certainty and never hide uncertainty, exclusions, or missing information.
2. Present information in simple language that patients and clinicians can understand.
3. Provide clear correction options when medication information or patient-entered information appears inaccurate.
4. Keep urgent-care guidance visually distinct from routine discussion points.
5. When uncertainty exists, default to transparency and clinician review rather than generating potentially misleading conclusions.

Uncertainty and Fallback:

Sufficient Information Available: When medication information is available in the catalogue and sufficient patient information has been provided, CarePulse generates the medication schedule, plain-language guidance, discussion prompts, and pre-consultation summary. Information is presented as *verified information* and is linked to validated reference sources.

Information Uncertain or Incomplete: When information is incomplete, uncertain, or cannot be confidently interpreted, CarePulse requests additional details, retains the patient's original wording, and clearly communicates the limitation. Discussion points may be presented as *possible discussion points*, while unsupported conclusions and rankings are withheld. Examples include incomplete information about a medication-related concern, uncertain interpretation of patient-entered information or missing details required to generate meaningful discussion prompts.

Failure or Fallback Behaviour: When CarePulse cannot safely generate a response, it defaults to transparency and clinician review rather than making assumptions.

Situation	**Fallback Response**
Unmapped medication	Request verification and exclude the medication from AI-supported analysis.

Situation	Fallback Response
Missing key information	Prompt the patient for additional details before continuing.
Uninterpretable patient concern	Retain the original entry and carry it forward to the consultation summary without AI-generated discussion points.
Potentially urgent concern	Bypass routine processing and provide urgent-care or clinician-review guidance.
AI-generated content fails safety checks	Suppress the output and replace it with validated reference information or static guidance.
AI-supported features unavailable	Continue providing the medication schedule and stored patient information.

Design Principle: When uncertainty exists, CarePulse should explain what is known, what is unknown, and what action the user should take next, rather than presenting potentially misleading conclusions.

Human-in-the-Loop:

CarePulse is designed to support communication and decision-making, not to replace clinical judgment. Human oversight is built into the workflow to ensure patient safety and maintain clinician authority.

Stage	Human involvement
Medication Selection	Patients select medications from a validated catalogue and can review entries for accuracy before use.
Medication Understanding	Patients review medication information and decide what concerns or questions they wish to discuss with their clinician.
Concern Documentation	Patients record medication-related concerns using their own observations and experiences.
Discussion Preparation	CarePulse organizes information and presents possible discussion points, but patients decide which concerns are relevant to raise during consultation.
Consultation Review	Clinicians review the pre-consultation summary alongside their clinical assessment and patient history.
Clinical Decision-Making	All diagnosis, treatment decisions, medication changes, and follow-up plans remain entirely under clinician authority.

Data, Model, and Evaluation Strategy:

Problem as an AI Task:

CarePulse is not designed to diagnose disease or recommend treatment. Instead, it uses AI to support three tasks:

1. *Information Simplification:* Converting complex medication information into patient-friendly language.
2. *Information Organization:* Structuring patient-reported concerns into meaningful categories.
3. *Information Summarization:* Generating concise pre-consultation summaries for clinician review.

This solution combines information retrieval, language understanding, and controlled text generation.

Data Sources:

External Data Sources:

- CDSCO-approved medication information
- Indian Pharmacopoeia Commission (IPC) resources
- Jan Aushadhi Medicine Information
- Validated drug monographs and medication safety references

These sources provide medication names, ingredients, indications, and safety information.

User-Generated Data:

- Selected medications
- Medication schedule
- Adherence information
- Patient-reported concerns and side effects
- Supplement use
- Consultation questions

Internal Data:

- Validated medication catalogue covering the supported MVP regimens
- Medication-to-ingredient mappings
- Safety rules and escalation criteria

Data Assumptions:

The MVP assumes that patients can identify the medications they are currently taking, supported medications exist within the validated catalogue and patients can provide basic information about concerns they experience.

Labelling and Ground Truth:

The MVP does not require large-scale machine learning training. Ground truth will be derived from validated medication reference sources, expert-reviewed medication mappings, predefined symptom categories and known medication safety information. Patient-entered information will be used for workflow testing rather than model training.

Baselines:

CarePulse will be evaluated against both existing workflows and simpler non-AI alternatives.

Current Workflow Baseline: evaluates whether CarePulse improves the overall patient-clinician process.

Workflow	Current Approach	CarePulse Approach
Medication management	Reminder applications and medication lists	Structured schedule with plain-language guidance
Concern documentation	Manual notes or memory	Organized concern tracking
Consultation preparation	Patient recall during consultation	Pre-consultation safety summary

AI Capability Baseline: evaluates whether AI-assisted features provide additional value beyond simpler rule-based or manual alternatives.

AI Capability	Baseline Comparator
Plain-language medication guidance	Fixed clinician-written medication descriptions
Concern organization	User-selected predefined categories only
Discussion prompts	List of known side effects without prioritization
Pre-consultation summary	Chronological log export without summarization
Red-flag detection	Rule-based keyword matching

Model or API Choice:

A hybrid Rules + Retrieval + Language Model approach is recommended for CarePulse. We shall use validated medication data and rules for safety-critical decisions, retrieval for trusted medication information, and language models for explanation, organization, and summarization.

Rule-based approach:

1. Medication catalogue and ingredient mapping - Medication information must be accurate, traceable, and based on validated reference sources.
2. Medication schedule generation - Scheduling logic is structured and predictable and does not require AI reasoning.

Retrieval-based approach: Medication information retrieval - Ensures explanations are grounded in validated medication information rather than generated from memory

Language Model + Rule-based approach: Discussion point generation - Creates structured discussion prompts while maintaining safety guardrails and appropriate uncertainty.

Language Model + Retrieval-based approach: Plain-language medication guidance - Helps translate complex medical information into language that patients can easily understand.

Language Model-based approach:

1. Supports interpretation and organization of patient-entered concerns using controlled terminology mappings.
2. Pre-consultation safety summary - Summarizes patient information into a concise format for clinician review.

Evaluation Framework:

Technical Metric	Threshold	Importance
Medication Mapping Accuracy	100% for supported medications	Any mapping error can affect all downstream outputs.
Medication Schedule Understanding	>=80% of patients correctly identify medication purpose and timing	Measures whether the schedule and guidance improve understanding.
Summary Accuracy and Completeness	>=95% of patient-entered information correctly represented in the summary	Ensures clinicians receive reliable information.

Business/Workflow Metric	Initial Threshold	Why It Matters
Consultation Preparedness	>=80% of patients report feeling better prepared for consultation	Primary measure of product value.
Medication Reconciliation Time	Reduction compared with current workflow	Measures operational efficiency.
Summary Usage Rate	>=80% of clinicians review the summary during simulated consultations	Indicates adoption and relevance.
Clinician Usefulness Rating	>=4/5 from at least 80% of reviewers	Measures usefulness and signal-to-noise ratio.

Trust & Safety Metrics	Initial Threshold	Why It Matters
Unsafe Medication Advice Rate	0%	The system must never recommend starting, stopping, or changing medications.
Red-Flag Escalation Accuracy	100% for predefined urgent concerns	High-risk concerns must always be escalated appropriately.
Unmapped Medication Rate	Monitored throughout testing	Indicates limitations of catalogue coverage.
Clinician Concern Rate	<20% of outputs flagged as misleading or irrelevant	Measures quality and relevance of generated outputs.
Guardrail Activation Rate	Monitored throughout testing	Confirms that safety controls remain active and effective.

Equity and Access Thresholds:

Equity / Economics Metric	Initial Threshold	Why It Matters
Task success by age band	No band more than 15 percentage points below the overall rate	The expected user base skews older. A product that works only for the youngest band has not met its own users.
Task success by digital familiarity	No group more than 15 percentage points below the overall rate	Low digital familiarity is a design problem to solve, not a user to exclude.
Task success by health literacy proxy	Monitored throughout testing; investigated if a gap exceeds 15 points	Reported as a proxy. Patients seek this product precisely because they are not clinically trained.
Red-flag escalation by language	100% per supported language, not pooled	A pooled figure can hide a language in which urgent expressions are missed.
Cost per enrolled patient-month	Measured and reported at pilot; no target set pre-pilot	Setting a target before measurement would be an unfounded claim.
Clinician minutes saved per consultation	Positive and measurable against the current-workflow baseline	If no time is saved, the institutional value proposition does not hold.

Subgroup reporting note: age bands and language are recorded directly. Digital familiarity and health literacy are short self-reported proxies collected at enrolment, not validated instruments, and are reported as such. A material gap triggers design investigation rather than exclusion of the affected group.

Test Design:

The MVP will be evaluated using a structured testing approach that combines functional, usability, and safety assessment. A predefined test set will include supported medication regimens, common medication-related concerns, misspelled medication names, duplicate products, supplements, ambiguous patient-reported concerns, and urgent scenarios. Outputs will be reviewed by independent reviewers to assess accuracy, usefulness, and potential safety concerns. AI-assisted workflows will be compared with baseline approaches such as static medication information, manual concern tracking, and non-summarized consultation records. Evaluation will consider differences across users with varying levels of age, medication complexity, health literacy, and digital familiarity. Testing will be repeated whenever significant changes are made to the medication catalogue, safety rules, prompts, or AI components.

Evaluation Principles: CarePulse shall be assessed not only on technical performance, but also on whether it improves medication understanding, supports safer documentation of concerns, and helps patients and clinicians communicate more effectively during consultations.

Launch Thresholds:

The MVP will be considered ready to progress to the next phase only if it meets predefined requirements for accuracy, safety, usability, and clinician usefulness.

Area	Threshold	Go/No-Go Rationale
Medication Mapping Accuracy	100% accuracy for all supported medications in the catalogue	Any medication mapping error can affect all downstream outputs and is therefore unacceptable.
Unsafe Medication Advice	0% occurrence of recommendations to start, stop, substitute, or change medications	CarePulse must remain an educational and communication tool.
Pre-Consultation Summary Accuracy	>=95% of patient-entered information correctly represented in the summary	Clinicians must be able to trust the information presented.
Medication Schedule Understanding	>=80% of patients correctly identify medication purpose and timing	Demonstrates that the schedule and guidance improve understanding.
Consultation Preparedness	>=80% of patients report feeling better prepared for consultations	Primary validation of the product value proposition.
Clinician Usefulness Rating	>=4/5 rating from at least 80% of clinicians reviewing the summary	Indicates that the summary has sufficient signal-to-noise ratio for practical use.

Area	Threshold	Go/No-Go Rationale
Red-Flag Escalation Accuracy	100% of predefined urgent concerns correctly escalated	Any failure may have patient-safety implications and is a launch blocker.
Guardrail Performance	No unsafe output reaches the user without suppression or intervention	Confirms that safety controls are functioning as intended.
Lay-to-Clinical Term Mapping	Mapping quality meets the agreed precision and recall targets against the reviewer-coded gold set, in every supported language	Symptom normalisation is where patient language enters the safety logic. A failure here propagates to every downstream output.
Subgroup Task Success	No reported age or digital-familiarity group more than 15 percentage points below the overall task success rate	A product intended for older adults managing complex regimens must work for them, not only for the most digitally confident users.

Go/No-Go Decision:

CarePulse should proceed from MVP evaluation to pilot preparation only if:

- Medication information is consistently accurate.
- No unsafe medication advice is generated.
- Patients demonstrate improved medication understanding and consultation preparedness.
- Clinicians find the pre-consultation summary useful in reviewing patient information.
- Safety controls reliably identify and escalate urgent concerns.

Launch Principle:

CarePulse should move forward only when it demonstrates meaningful value for patients and clinicians while maintaining strict medication-safety boundaries. Safety and accuracy take precedence over feature completeness.

Business, Economics, and Scaling:

Monetization Model: Recommended monetization model is Enterprise/Clinic Subscription. The primary users of CarePulse are patients and clinicians, but the most likely paying customers are hospitals, diabetes clinics, and chronic disease management programs.

Customer	Value Provided	Monetization Approach	Price point
Hospitals and diabetes clinics	Improved consultation preparation and medication reconciliation.	Annual institutional subscription, charged per enrolled patient per month.	Proposed test price: Rs 179 per patient-month (150 to 999 patients)
Chronic disease management programmes	Better patient engagement and structured medication support.	Per enrolled patient per month on an annual contract. The budget line is the clinical programme, not IT.	Proposed test price: Rs 158 per patient-month (1,000 to 9,999)
Hospital chains and enterprise	Scalable medication-management workflow across sites, with custom catalogue additions.	Enterprise agreement with an annual governance review.	Proposed test price: Rs 132 per patient-month (10,000+)
Pilot sites	Supervised evaluation with full support and no production claims.	Fixed programme fee rather than a per-patient price, set to recover running cost.	Proposed test price: Rs 4,50,000 for 100 patients over 6 months
Individual patients	Access to medication support and consultation preparation.	Free in every tier. Charging patients directly for a medication-safety tool would leave the people at highest risk least likely to have it.	Rs 0

Why This Model?

The primary value is created during the clinical encounter. Clinicians and healthcare organizations benefit from improved workflow efficiency. Patients are often reluctant to pay directly for medication-management tools. Institutional adoption supports broader and more sustainable deployment.

Competitive Price Benchmarking

Pricing was framed using adjacent budget benchmarks and then tested against the internal cost model. The external products are not direct feature-level substitutes for CarePulse; they indicate the per-patient budgets used for diabetes-management support. Two different comparator sets apply, and they give opposite answers. Which one is relevant depends entirely on whose budget the product is bought from, and that makes the choice of buyer a pricing decision rather than a sales decision.

Budget benchmark A: adjacent diabetes-management programmes

Product	Price	Per patient-month	Who pays	Note
Fitterfly Diabetes Prime, 12 months	Rs 39,999 list Rs 19,999 launch	Rs 3,333 list Rs 1,667 launch	Patient	Digital therapeutic with coaching. CGM reader not included.
BeatO Comprehensive Diabetes Care	Rs 1,500 to 3,000 per month	Rs 1,500 to 3,000	Patient	Coaching plus connected glucometer. Test strips charged separately.
mySugr Pro	Rs 800 per month	Rs 800	Patient	Logging and reporting app. Free tier available.
CarePulse, proposed	Rs 1,898 per patient-year	Rs 158	Institution	Proposed institutional test price. CarePulse has a narrower scope than the adjacent programmes, and the patient pays nothing.

This benchmark shows that the proposed CarePulse price is below broader patient-paid diabetes programmes. It does not establish feature equivalence or willingness to pay. CarePulse is a narrower medication-understanding and consultation-preparation layer purchased by an institution, so the proposed price must be tested with pilot buyers.

IT budget benchmark B: clinic and hospital software, not a competitive product comparison

Market anchor	Price	Unit	Note
e-Sushrut Clinic HMIS	Rs 299 to 499	per facility-month	Government HMIS through NHA and C-DAC. Rs 499 list, Rs 299 after subsidy.
Basic clinic EMR or SaaS	Rs 500 to 1,000	per facility-month	Indian EMR market survey.
Comprehensive clinic software	Rs 800 to 5,000	per facility-month	Indian EMR market survey.
Enterprise hospital systems	Rs 5,000 to 15,000	per facility-month	Indian EMR market survey.
CarePulse at a 500-patient clinic	Rs 79,087	per facility-month	5.3 times the top of the enterprise anchor, which buys an entire hospital system rather than one medication companion.

Against this set CarePulse is unaffordable, and not by a margin that discounting can close. Discounting a 500-patient clinic to Rs 15,000 per month would produce a contribution of minus Rs 12,000 per month, so every additional customer of that kind would deepen the loss.

The two comparisons are not in conflict. They describe two different budgets. Sold as an IT line item against facility software, the price is indefensible. Sold as a chronic-care programme against per-patient diabetes programmes, it is cheap. The pricing recommendation follows directly: charge per enrolled patient per month, and sell only to buyers with a clinical-programme budget. Approaching an IT budget with this price would fail on the first comparison a buyer makes.

The proposed test price of Rs 158 per patient-month produces an approximately 50% contribution margin against the included modelled steady-state cost of Rs 79. A 70% margin, normal for software, is not reachable at this cost structure, because the cost is clinical review and human support rather than compute. The price should not be published before the pilot measures M-21 and M-24, since the dominant input, support cost at Rs 30 per patient-month, is currently an assumption.

Main Trade-Offs

Option	Advantages	Limitations
Direct patient subscription	Simple to launch	May limit adoption and accessibility
Usage-based pricing	Matches activity levels	More complex to administer and predict

Option	Advantages	Limitations
Enterprise subscription	Aligns with clinical workflow and organizational value	Requires institutional adoption and stakeholder buy-in
Hybrid model	Multiple revenue streams	Operationally more complex

Value Logic:

CarePulse creates value for patients, clinicians, and healthcare organizations by improving the flow of medication-related information between home and clinic.

- *Patient value:* Better understanding of medications, structured recording of concerns, and improved preparation for consultations.
- *Clinician value:* Less time spent reconstructing medication history and reviewing fragmented information during follow-up visits.
- *Healthcare Provider value:* More efficient consultations, improved continuity of care, and a differentiated chronic disease support service.
- *Organization value*: A reusable workflow that could potentially be extended to other chronic conditions involving complex medication management.

Value Proposition: CarePulse creates value by helping patients understand their medication plan, document concerns between visits, and arrive at consultations with better organized information for clinician review.

Cost Drivers:

Cost Area	MVP Approach	Pilot Rs / month	Steady state Rs / month	Scale Implication
Medication catalogue management	Manual validation of supported regimens, with second-reviewer approval.	3,000	25,000	Driven by regimen count, not patient count. Rs 6,000 of reviewer time per regimen added (M-24).
AI and API usage	Constrained symptom normalisation and template-first summary generation.	9	434	Scales with usage and is immaterial. Rs 0.09 per patient-month, 0.1% of running cost. A frontier model would multiply this by roughly twenty and still not matter.
Clinical governance	Medical advisor time for safety rules, escalation criteria and incident review.	24,000	60,000	Rises in steps as sites are added, not linearly with patients.

Cost Area	MVP Approach	Pilot Rs / month	Steady state Rs / month	Scale Implication
User support	Self-service onboarding with assisted help where needed.	20,000	1,50,000	The dominant variable cost at Rs 30 per patient-month, and the only line product design can move. Reducing it is a product objective, not an operations one.
Cloud hosting	Managed database, application hosting, storage and backups.	25,000	1,20,000	Broadly proportional to patients, at Rs 24 per patient-month.
Privacy, security and compliance	Basic controls during MVP testing. External review before any real patient data.	0	40,000	Rs 4,00,000 one-time for external review and penetration test, then a fixed ongoing obligation once live.
MedDRA licence	Terminology target for the mapping layer. Eligibility for a no-fee subscription must be confirmed with MSSO for the relevant subscriber category and use.	0	0	Commercial terms are not assumed. Eligibility must be confirmed before pilot and reassessed before commercialisation; any fee enters fixed cost.
Total per month		72,009	3,95,433	Cost per patient-month falls from Rs 1,304 at 100 patients to Rs 79 at 5,000 and flattens near Rs 58. The floor is set by support and cloud, not by the model.

The finding worth stating is that inference is 0.1% of running cost. Clinical review, support and governance are the cost of this product. Model selection is therefore a quality and latency decision rather than a cost decision, and any plan that treats token price as the lever is optimising the wrong line.

Unit Economics:

At the MVP stage, the objective is to demonstrate value rather than profitability. Long-term sustainability depends on whether the value created exceeds the operating cost of maintaining the platform.

Monthly economic value estimate per enrolled patient = clinician reconciliation minutes saved x loaded clinician cost + avoidable support contacts reduced x contact cost + retention/engagement value - monthly technology, review, support, and compliance cost. Time saved becomes a cash benefit only if the organization can redeploy capacity or accommodate additional consultations. No financial savings claims should be made until they are demonstrated through pilot testing.

Unit Economics Principle: The workflow should create more value through improved consultation efficiency and patient support than it consumes in technology and operational costs.

Viability, Feasibility and Monetizability Assessment

Feasibility: CarePulse is technically feasible within a controlled scope because medication facts, schedules, unsupported-medicine handling and fixed safety messages are separated from language-model tasks. The main execution challenge is maintaining accurate Indian product data, multilingual terminology and clinical governance.

Viability: The problem and workflow are credible, but viability is not yet proven. It depends on patients completing the workflow, clinicians using the snapshot, and a measurable reduction in medication-reconciliation effort without creating additional burden. These assumptions will be tested during usability evaluation and the controlled pilot.

Monetizability: CarePulse has a plausible institutional revenue model through hospitals and chronic-disease programmes on a per-enrolled-patient basis. The proposed price remains a test hypothesis until pilot buyers confirm willingness to pay and actual support, governance and operating costs are measured.

Decision: Proceed to evaluation MVP development. Do not proceed to a real-patient pilot or scale until the safety, workflow and commercial gates are met.

Scaling Constraints and Risks:

As CarePulse expands beyond the MVP, several operational and technical challenges must be addressed.

- *Medication catalogue growth:* Additional medications, brands, and fixed-dose combinations will require ongoing validation and maintenance.
- *Regional medication variation:* Differences in local formulations and brand names increase medication-mapping complexity.
- *Language support:* Expanding into multiple languages requires careful review to preserve medical accuracy and readability.

- *Cost growth:* AI usage, medication catalogue maintenance, clinical review, user support, and compliance activities may increase operating costs as adoption grows.
- *Workflow complexity:* Supporting larger patient populations, more medication regimens, and additional use cases may increase product complexity and reduce usability if not carefully managed.
- *Governance burden:* Medication information, safety rules, AI outputs, and clinical risk controls require continuous monitoring, review, and updating.
- *Regulatory considerations:* Expanding the intended use toward diagnosis or treatment recommendations would substantially increase regulatory requirements and liability exposure.
- *System integration:* Future integration with electronic health records or hospital systems may create additional technical and operational challenges.
- *Latency and performance:* As usage grows, response times for AI-supported features and summary generation must remain acceptable for routine clinical use.
- *Alert fatigue:* Urgent escalation rules must maintain high sensitivity without generating excessive alerts that reduce clinician trust.

Scaling Principle:

CarePulse should scale only when medication accuracy, safety, usability, and governance can be maintained at the same standard achieved during the MVP and pilot stages. Growth should not compromise patient safety, clinician trust, or workflow simplicity.

Ethics, Governance, and Risk Note:

Main Risks:

1. *Unsafe or Misleading Outputs:* Patients may misunderstand discussion points as diagnosis, treatment advice, or confirmation of medication-related effects.
2. *Medication Information Errors*: Incorrect medication mapping or unsupported medications may lead to inaccurate guidance.
3. *Privacy and Data Protection:* Medication information, patient concerns, and consultation summaries contain sensitive health information that must be handled securely.
4. *Patient Over-Reliance*: Patients may treat the absence of a warning as reassurance and delay seeking medical advice.
5. *Clinician Adoption Risk*: If the consultation summary contains too much irrelevant information, clinicians may stop using it.

Harm Scenarios:

Potentially Affected Group	Possible Harm
Patients	Misinterpretation of discussion prompts as medical advice, delayed care, or inappropriate medication decisions.

Potentially Affected Group	Possible Harm
Clinicians	Reliance on incomplete, inaccurate, or poorly summarized information during consultation.
Healthcare Organizations	Reputational, legal, or regulatory consequences if unsafe outputs reach users.

Risk Register:

Risk	Potential harm	Control	Launch blocker
Wrong ingredient mapping	Incorrect schedule or discussion	Validated catalogue; no free-form inference	Any unresolved supported-catalogue error
Overstated causality	Patient stops medicine	Possible-not-proven language; no action advice	Any unsafe instruction
Missed urgent concern	Delayed care	High-recall screen; urgent escalation copy	Sensitivity below threshold
Hallucinated snapshot	Clinician distracted or misled	Template-first summary; claim traceability	Unsupported statement rate above threshold
Privacy breach	Exposure of health information	Data minimization, encryption, access controls	Critical security issue
Bias / accessibility	Lower benefit or higher error for subgroups	Subgroup testing and language review	Material disparity without mitigation
Automation bias	Blind trust in output	Reason display, uncertainty labels, override capture	Clinician review workflow absent
Scope creep	Unvalidated clinical claims	Intended-use governance and change control	Product begins recommending treatment

Controls:

Bias and Fairness: Evaluate usability across different age groups, literacy levels, medication complexity, and levels of digital familiarity. Ensure that unsupported medications and missing information are handled consistently rather than inferred.

Safety Guardrails:

1. No diagnosis, treatment recommendations, or medication-change instructions.
2. Medication information limited to the validated catalogue.
3. Red-flag concerns are routed to clinician review or urgent-care guidance.
4. Unsupported medications are excluded from AI-supported analysis.

Privacy Controls:

1. Collect only the information required to support the workflow.
2. Separate patient identity from usage data wherever possible.
3. Apply appropriate security controls before any use of real patient information.

Review and Escalation:

1. Medication mappings and safety rules are validated before inclusion.
2. High-risk concerns bypass routine AI processing and trigger escalation guidance.
3. Clinicians remain responsible for all treatment decisions.

Logging and Traceability:

1. Medication information and summaries remain linked to validated reference sources.
2. System actions, safety-rule activations, and fallback decisions are traceable and reviewable.

Launch Blockers:

CarePulse should not proceed to pilot deployment if any of the following are identified:

1. Medication mapping errors within the supported catalogue.
2. Any output recommending medication initiation, discontinuation, substitution, or dose modification.
3. Failure to escalate predefined urgent concerns appropriately.

4. Significant inaccuracies in the pre-consultation summary.
5. Inability to clearly communicate uncertainty, limitations, or unsupported medications.
6. Privacy, security, or governance requirements not being met.

Governance Principle: CarePulse is intended to support medication understanding, communication, and consultation preparation. Patient safety, transparency, privacy, and clinician oversight take priority over feature expansion or deployment speed.

Governance roles: Patients and treating clinicians are the product users. A clinical pharmacist prepares product and ingredient records; an independent clinical reviewer approves activation; a pharmacovigilance reviewer owns adverse-reaction terminology and MedDRA mappings; and the clinical owner approves urgent-expression criteria, safety policy and incident response. These roles maintain the product content and do not replace the treating clinician's patient-specific judgement.

Role of Team Members:

Team Members	BITS ID	Role in Project Work
Ramyatha Munnangi	2025MB26036	Product Strategy and Research
Aishwarya M	2025MB26172	Clinical Workflow and Governance
Subhasree Dutta	2025MB26011	UX and Prototype
Anushka Tripathi	2025MB26168	PRD and Evaluation
Sumathi Senthil Kumar	2025MB26068	Business and Scaling

Each team member prepared and developed the sections related to their interest or domain following a team meeting to agree upon division of work. We had regular discussions, check-ins and comments wherein team members cleared doubts and suggested improvements. Drafts prepared by one team member were reviewed by others, and feedback was incorporated before finalizing the artifacts. Important decisions were discussed as a group and finalized after the entire team was ok with it.

AI Collaboration Log:

Tool: Claude (Anthropic)

Used for: Coding the functional prototype from the PM artifact, drafting the symptom-terminology design, researching free drug-reference APIs, compiling and verifying the reference list, document formatting.

Sample Prompt:

Build a functional browser-based prototype of CarePulse, the AI diabetes medication safety and consultation companion specified in our PM artifact. Use only what the document specifies. Do not invent clinical content.

SCOPE - build Module 2 and Module 3 end to end, plus the Module 1 schedule and plain language guide. Patient view and clinician view.

ARCHITECTURE - follow the hybrid stated in "Model or API Choice": Rules for the medication catalogue, ingredient mapping and schedule. Retrieval from validated catalogue records for medication guidance. Language model only for symptom normalization, organizing concerns and summarizing. Simulate the language steps with a scripted engine so the demo runs without an API key, and declare that on the page.

CATALOGUE - six curated regimens as a subset of the 15 regimen patterns in our MVP catalogue table, each with brand, active ingredients and strengths, timing, purpose, plain language guidance and label-derived effect profile.

MUST DEMONSTRATE the four fallback behaviours from our "Uncertainty and Fallback" table, each as a scripted path: Unmapped medication: record, flag for verification, exclude from all AI-supported analysis, Missing key information: prompt for detail, Uninterpretable concern: retain original wording verbatim, generate nothing, carry forward to the clinician, Potentially urgent concern: bypass routine processing before the language, step, show fixed urgent-care guidance, name no medicine as a cause

SAFETY CHECKS - deterministic rules only, no model involvement: identify the same active ingredient appearing in more than one entered product. State clearly that the check covers only entered medicines and is not a complete interaction or dose-safety screen. Do not prioritise a symptom based on molecule overlap. Urgent escalation must arise only from the reviewed urgent-expression pathway.

ADVERSE EVENT ATTRIBUTION - where a normalized symptom appears in more than one medicine's effect profile, show every candidate with the date each was started. Do not rank, score or select a cause. Causality assessment is out of scope per our PRD.

GUARDRAILS from our Risk Register and Safety Guardrails: No output may recommend starting, stopping, substituting or changing a dose. Screen generated text for this language and suppress on a match. Possible-not-proven language on every discussion point. Uncertainty labels: Verified Information, Possible Discussion Point, Insufficient Information. Hide confidence scores that create false precision, per our "What should be hidden" list

TRACEABILITY - every output carries a visible trace naming which layer produced or blocked it: validated catalogue, red-flag screen, language step, retrieval, output guardrail.

CLINICIAN VIEW - the pre-consultation snapshot, a relevant / not relevant control on each discussion point to capture override, and a live metrics panel reading the thresholds in our Evaluation Framework.

CONSTRAINTS - single self-contained HTML file, no server, no account, synthetic data only, works offline. State on the page what the prototype does not prove.

Human validation: The prototype code and generated design suggestions were checked against the PRD, intended-use boundary, safety rules and scripted test paths. Product scope, risk acceptance, metric thresholds, pricing assumptions and the final recommendation were decided by the team.

References:

Sources cited in the Business Problem

1. Flood D, Green H, Hu P, Ali MK, Shete A, Pedgaonkar S, Langa KM, Crimmins EM, Bloom DE, Sekher TV, Lee J. Prevalence, awareness, treatment, and control of diabetes in India: a nationally representative survey of adults aged 45 years and older. The Lancet Global Health. 2025;13(9):e1543-e1552. doi:10.1016/S2214-109X(23)00502-8
2. Dalal J, Chandra P, Chawla R, Kumar V, Abdullakutty J, Natarajan V, Naqvi SMH, Gaurav K, Rathod R, Dhanaki G, Kotak B, Shah S. Clinical and Demographic Characteristics of Patients with Coexistent Hypertension, Type 2 Diabetes Mellitus, and Dyslipidemia: A Retrospective Study from India. Drugs - Real World Outcomes. 2023;11(1):167-176.
3. Adepu SK, Sirasu M, Ravinder G. Impact of polypharmacy in older patients with type 2 diabetes in rural population in South India. Asian Journal of Medical Sciences. 2022;13(10):108-111. doi:10.3126/ajms.v13i10.47279
4. Pathak N, Singh S, Sachan AK, Usman K, Sindhu S. A Study on Adverse Drug Reactions in Type 2 Diabetes Mellitus Patients on Oral Anti-Diabetic Drugs in a Tertiary Care Hospital in Northern India. International Journal of Life Sciences Biotechnology and Pharma Research. 2024.
5. Huang Z, Lum E, Jimenez G, Semwal M, Sloot P, Car J. Medication management support in diabetes: a systematic assessment of diabetes self-management apps. BMC Medicine. 2019;17:127. doi:10.1186/s12916-019-1362-1

Sources supporting the 15-regimen MVP catalogue

The catalogue scope was derived from an Embase literature search of Indian real-world studies, prescribing-pattern studies and guidelines, limited to 2000-2026 and to case reports, Cochrane reviews, controlled clinical trials, literature reviews, meta-analyses, observational studies,

randomised controlled trials and systematic reviews. The search identified regimen patterns in use in India together with the medication-related concerns reported alongside them. The principal sources are listed below; the full search strategy and complete citation set are retained in the project record.

6. Jadhav M, Fernandes M, Choudhari S, Bhushan S, et al. A real-world study on drug utilization of fixed dose combination of teneligliptin + pioglitazone + metformin in Indian patients with type 2 diabetes mellitus. International Journal of Diabetes in Developing Countries. 2024.

7. Raju GA, Ganesh HK, Anakal M, Reddy SVB, et al. Effectiveness of glimepiride and metformin in managing long-term type 2 diabetes in the Indian population. Asian Journal of Diabetology. 2025.

8. Rajsantan K, Balaji T, Sharmeela S. A prospective, observational study to understand the drug usage composition of anti-diabetic medications in a tertiary care hospital. Journal of Cardiovascular Disease Research. 2024.

9. Karanth JB, Maribashetti K, Karanth GJ. Clinical profile, treatment combinations, and glycemic outcomes in Indian patients with type 2 diabetes: an observational study. National Board of Examinations Journal of Medical Sciences. 2026.

10. Ghosh S, Tiwaskar M, Chawla R, Jaggi S, et al. Teneligliptin real-world effectiveness assessment in patients with type 2 diabetes mellitus in India: a retrospective analysis (TREAT-INDIA 2). Diabetes Therapy. 2020.

11. Kumar V, Agarwal S, Saboo B, Makkar B. RSSDI guidelines for the management of hypertension in patients with diabetes mellitus. International Journal of Diabetes in Developing Countries. 2022.

12. Gour PR, Bagati KD, Balakrishnan S, Chitti Babu G, Ramana Murty Kadali SLDV. Trends and management of comorbid conditions in diabetes mellitus type 2 patients in a tertiary care hospital. International Journal of Research in Pharmaceutical Sciences. 2020.

13. Rahaman SKS, Biswas RK, Pal A. Adherence to American Diabetes Association and National List of Essential Medicines 2022 guidelines in antidiabetic prescriptions: a cross-sectional study from Eastern India. Bengal Physician Journal. 2026.

14. Kumar L, Kumar P, Kumari A. A study on the prescribing pattern of antidiabetic drugs in patients with type 2 diabetes mellitus. International Journal of Pharmaceutical Quality Assurance. 2025.

15. Das AK, Mithal A, Joshi S, Kumar KMP, et al. Baseline characteristics of participants in the LANDMARC trial: a 3-year, pan-India, prospective, longitudinal study to assess management and real-world outcomes of diabetes mellitus. Endocrinology, Diabetes and Metabolism. 2021.

Reference data sources used by the product

16. Central Drugs Standard Control Organisation (CDSCO), Government of India. Approved product information.

17. Indian Pharmacopoeia Commission (IPC). Monographs and Pharmacovigilance Programme of India safety communications.

18. Pharmaceuticals and Medical Devices Bureau of India. Jan Aushadhi product list.

19. US National Library of Medicine. RxNorm and the RxNav application programming interface.

20. US Food and Drug Administration. openFDA drug label application programming interface.

21. WHO Collaborating Centre for Drug Statistics Methodology. Anatomical Therapeutic Chemical (ATC) classification and Defined Daily Dose (DDD) index.

22. International Council for Harmonisation. Medical Dictionary for Regulatory Activities (MedDRA), maintained by the MedDRA Maintenance and Support Services Organization.